CHAPTER 76

Male reproductive system

INTRODUCTION

The male reproductive system consists of the gonads (testes), spermatic cord, sex accessory glands and external genitalia. The testes perform both spermatogenic and steroidogenic functions. The unique anatomy of the reproductive tract, both gross and microscopic, is optimally suited for these functions to be carried out efficiently and effectively.

TESTIS AND EPIDIDYMIS

TESTIS

The testes are ovoid organs responsible for sperm and testosterone production. In adults, they typically measure 4–5 cm in length, 2–3 cm in breadth and 3–4 cm in anteroposterior diameter (Tishler 1971); their weight varies between 12 and 20 g. Average testicular volume ranges from 15 to 25 ml (Prader 1966, Goede 2011).

The testes are suspended in the scrotum by the spermatic cord (**Figs 76.2**, **76.3**). The left testis usually lies lower than the right testis; both are positioned obliquely, such that the upper pole is tilted anterolaterally while the lower pole is tilted anteromedially. The location of the testes in the scrotum, combined with the characteristics of the scrotal skin, as well as the counter-current heat exchange mechanism of the testicular pampiniform plexus, maintains the testes at a temperature 3–4°C below body temperature.

The testes are enclosed in a tough capsule made up of three layers: an innermost tunica vasculosa, an intermediate tunica albuginea and an outer tunica vaginalis (**Fig. 76.4**). The posterior aspect of the testis is the site of attachment of the epididymis and is, therefore, only partly covered by serosa. Here, the tunica albuginea projects inwards to form the mediastinum testis, and the tunica vaginalis projects outwards to cover the epididymis. Within the scrotum, the testes are separated by a fibrous median septum.

Tunica vasculosa The tunica vasculosa contains a plexus of blood vessels and loose connective tissue. It lines the inner surface of the tunica albuginea, as well as all of the surfaces and septations within the testis.

Tunica albuginea The tunica albuginea is a dense, blue–white layer composed primarily of collagen fibres. It covers the tunica vasculosa and is surrounded by the visceral layer of the tunica vaginalis. At the posterior aspect of the testis, it projects inwards as a thick but incomplete fibrous septum – the mediastinum testis – which extends from the upper to the lower pole of the testis. It is here that vessels, nerves and testicular ducts traverse the testicular capsule (**Fig. 76.5**).

Tunica vaginalis The tunica vaginalis is a continuation of the peritoneal processus vaginalis, whose formation precedes the descent of the fetal testis from the abdomen to the scrotum. Following testicular migration into the scrotum, the portion of the processus vaginalis that falls between the internal inguinal ring and the testis contracts and is obliterated, leaving a distal sac containing the testis. Failure to obliterate the processus vaginalis results in a persistent communication with the scrotum and peritoneal cavity (**Fig. 76.6**), which can lead to hydroceles and indirect inguinal hernias.

The tunica vaginalis is reflected from the surface of the testis on to the inner surface of the scrotum, forming visceral and parietal layers that are continuous at both poles of the testis. The visceral layer covers all aspects of the testis, except the posterior aspect, where it is reflected towards the epididymis before becoming continuous with the parietal layer. The inner surface of the parietal layer has a smooth, moist, mesothelium. The potential space between the visceral and parietal layers – the cavity of the tunica vaginalis – is normally occupied by a thin film of clear, straw-coloured fluid. The volume of this fluid can increase with obstruction of lymphatic drainage, due to inflammatory, traumatic or neoplastic conditions of the testis, resulting in a hydrocele.

Vascular supply and lymphatic drainage

Arteries

The arterial supply to the testis and epididymis is derived from three sources. In descending order of contribution, these are the testicular artery (supplying approximately two-thirds of the testicular blood supply), the vasal artery (artery to vas deferens, artery to ductus deferens, vasal artery, ductal artery) and the cremasteric arteries (together supplying approximately one-third of the testicular blood supply) (**Fig. 76.7**) (Harrison and Barclay 1948, Raman and Goldstein 2004).

Testicular artery

The testicular artery (internal spermatic artery) arises from the abdominal aorta, inferior to the origin of the renal artery, and courses inferolaterally under the parietal peritoneum, along psoas major, towards the pelvis. On the right, it courses anterior to the inferior vena cava and posterior to the middle colic and ileocolic arteries and the terminal ileum. On the left, it courses posterior to the inferior mesenteric vein, left colic artery and the descending colon. As the right and left testicular arteries enter the pelvis, they lie anterior to the genitofemoral nerves, ureters and external iliac arteries. Both arteries then enter the deep internal inguinal ring and travel with the ipsilateral spermatic cord in the inguinal canal to the scrotum (see Figs 76.2, 76.5).

In its course to the testis, the testicular artery gives off one or more internal spermatic arteries, an inferior testicular artery, and branches supplying the caput, corpus and cauda epididymis (Macmillan 1954). The level at which this branching occurs is variable; in 31–88% of cases, it occurs within the inguinal canal (Beck et al 1992, Jarow et al 1992). At the level of the testis, branches of the testicular artery enter the tunica albuginea in the mediastinum testis and ramify in the tunica vasculosa before reaching their distribution. Ramification of the testicular arteries occurs primarily in the anterior, medial and lateral portions of the lower pole of the testis, and in the anterior segment of the upper pole (Jarow 1991), which has important implications for planning testicular biopsies. (For further reading about variations in the origin, course and number of the testicular arteries, see Asala et al (2001), Pai et al (2008)).

Vasal artery (artery to vas deferens, artery to ductus deferens, deferential artery, ductal artery)

The vasal artery is a branch of the superior (and, occasionally, inferior) vesical artery, which arises from the internal iliac artery.

Cremasteric artery

The cremasteric artery (external spermatic artery) is a branch of the inferior epigastric artery. It accompanies the spermatic cord and supplies the cremaster and other coverings of the cord. Both the vasal and the cremasteric arteries enter the inguinal canal at the deep inguinal ring, and travel the length of the spermatic cord alongside the testicular artery. The testicular artery and internal spermatic veins lie within the internal spermatic fascia, whereas the vas (ductus) deferens and its vessels, as well as the cremaster muscles and its vessels, lie outside the internal spermatic fascia but within the external spermatic fascia. In the scrotum, a rich vascular anastomosis occurs at the head of the epididymis, between the testicular and epididymal arteries, and at the tail of the epididymis between the testicular, epididymal, cremasteric and vasal arteries.

The volume of the left and right testis, as measured by ultrasonography in boys between infancy and adolescence, is shown in Table 76.1; reference curves for mean testicular volume are shown in Figure 76.1.

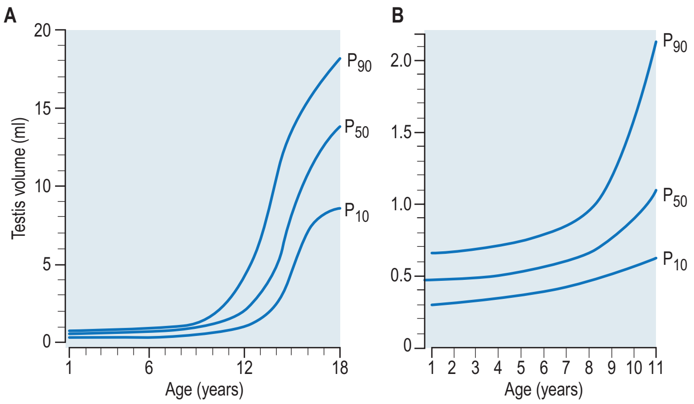

Fig. 76.1 A, Reference curves for mean testicular volume measured by ultrasound. P_{10}, P_{50} and P_{90} indicate tenth, fiftieth and ninetieth centiles, respectively. **B**, Enlargement of the reference curves for mean testicular volume measured by ultrasound. (With permission from Goede J, Hack WW, Sijstermans K et al; Normative values for testicular volume measured by ultrasonography in a normal population from infancy to adolescence. Horm Res Paediatr. 2011;76(1):56–64.)

Table 76.1 Volume of the left and right testis, as measured by ultrasound in boys between infancy and adolescence

		Ultrasound (ml)					
Age (years)	Boys (n)	Left volume	SD	Right volume	SD	Mean volume	SD
1	40	0.48	0.14	0.48	0.13	0.48	0.13
2	38	0.47	0.09	0.45	0.1	0.46	0.09
3	36	0.52	0.18	0.5	0.13	0.51	0.15
4	38	0.52	0.18	0.5	0.15	0.51	0.16
5	48	0.59	0.15	0.58	0.15	0.58	0.15
6	42	0.63	0.25	0.64	0.28	0.63	0.26
7	62	0.64	0.18	0.66	0.18	0.65	0.17
8	59	0.64	0.2	0.67	0.24	0.66	0.22
9	53	0.78	0.46	0.8	0.48	0.79	0.46
10	49	0.95	0.51	0.99	0.52	0.97	0.51
11	60	1.31	0.95	1.35	1.14	1.33	1.03
12	55	2.31	1.8	2.35	1.79	2.33	1.77
13	47	4.21	2.44	4.62	2.95	4.42	2.66
14	35	7.2	4.13	7.42	4.16	7.31	4.11
15	26	8.69	3.06	8.69	2.86	8.69	2.91
16	31	11.48	2.99	11.55	3.24	11.51	3.03
17	27	12.14	2.87	12.09	2.95	12.12	2.8
18	23	13.67	3.49	13.8	3.77	13.73	3.51

(SD = standard deviation)
(With permission from Goede J, Hack WW, Sijstermans K et al; Normative values for testicular volume measured by ultrasonography in a normal population from infancy to adolescence. Horm Res Paediatr. 2011;76(1):56–64.)

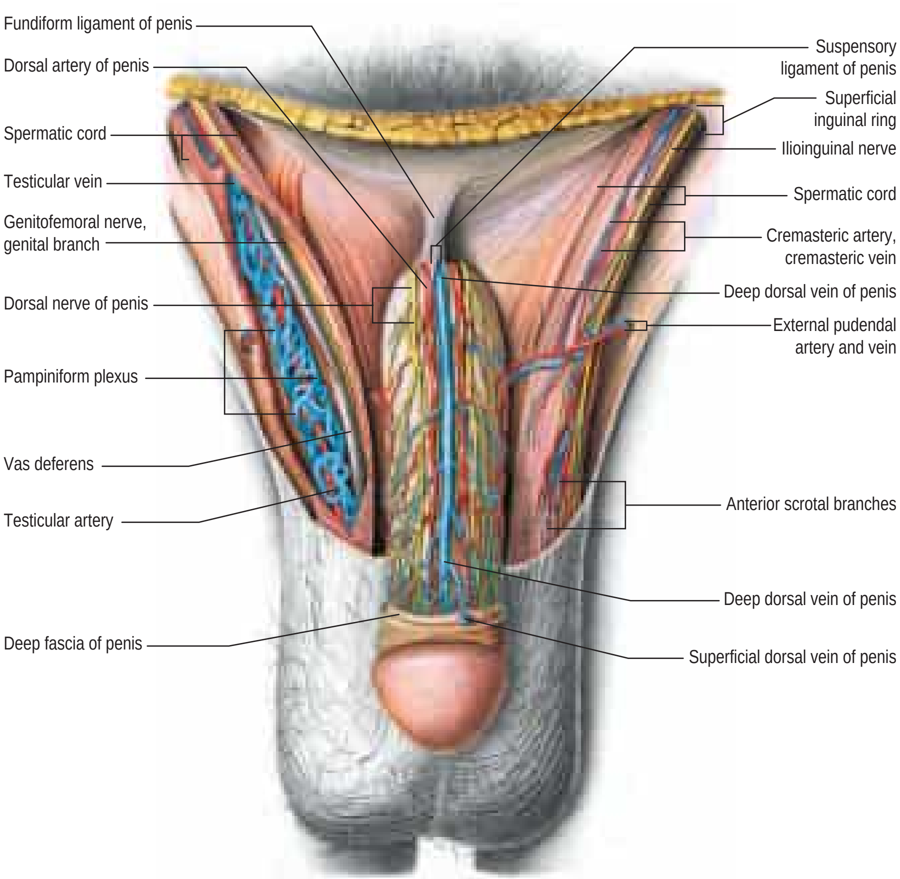

Fig. 76.2 The external male genitalia, ventral aspect. Nerves and vessels exposed by extensive removal of the skin and the superficial fascia of the penis. The layers of the spermatic cord have been incised on the right; note the pampiniform venous plexus surrounding the testicular artery. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban and Fischer. Copyright 2013.)

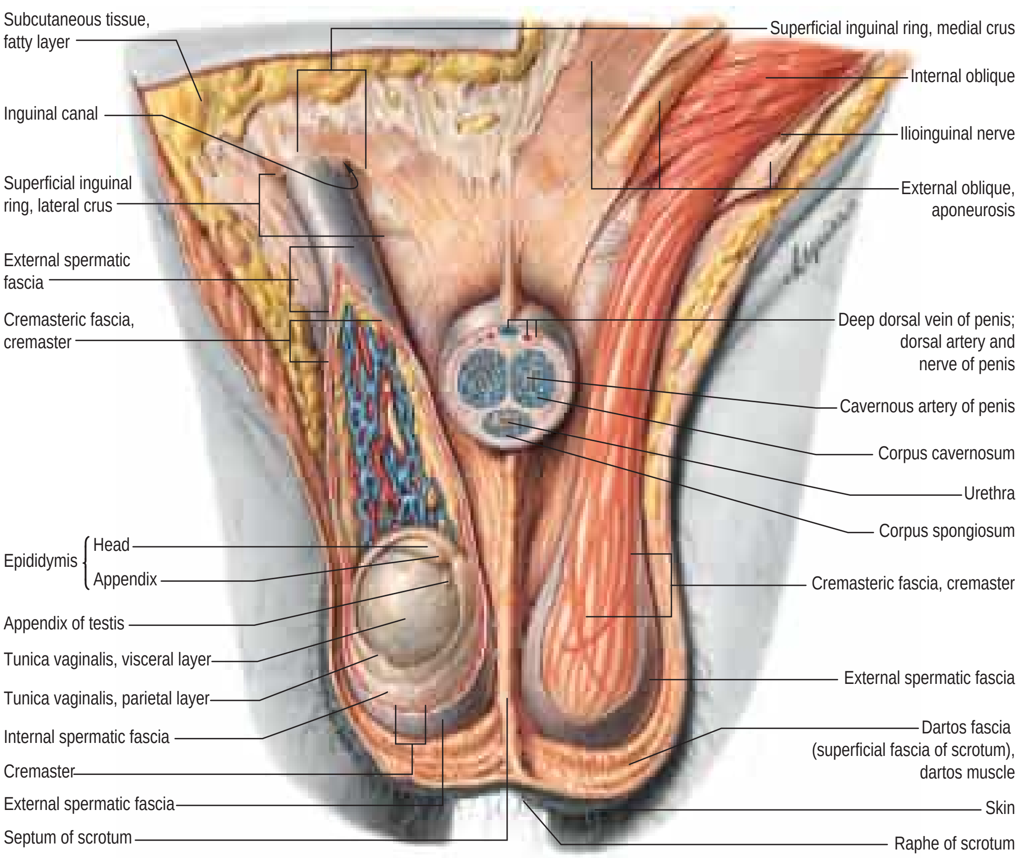

Fig. 76.3 The external male genitalia, ventral aspect. The skin of the abdomen and parts of the skin of the scrotum have been removed, and the body of the penis has been severed, revealing the internal structure of the penis. The layers of the spermatic cord and the coverings of the testis have been dissected on the right. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban and Fischer. Copyright 2013.)

Veins

Pampiniform plexus

Testicular veins emerge posteriorly from the testis, drain the epididymis, and unite to form several highly anastomotic channels surrounding the testis, known as the pampiniform plexus, a major component of the spermatic cord (see Fig. 76.2). This vascular arrangement means that counterflowing arteries and veins are separated only by the thickness of their vascular walls, permitting the exchange of heat and small molecules (Harrison 1949), and facilitating the maintenance of lower testicular temperatures. The pampiniform plexus ascends anterior to the vas deferens, and is drained by 3–4 veins in the inguinal canal. The veins enter the abdomen through the deep inguinal ring and coalesce into a single testicular vein that drains into the inferior vena cava on the right, and into the renal vein on the left. The right testicular vein joins the inferior vena cava at an acute angle, just inferior to the level

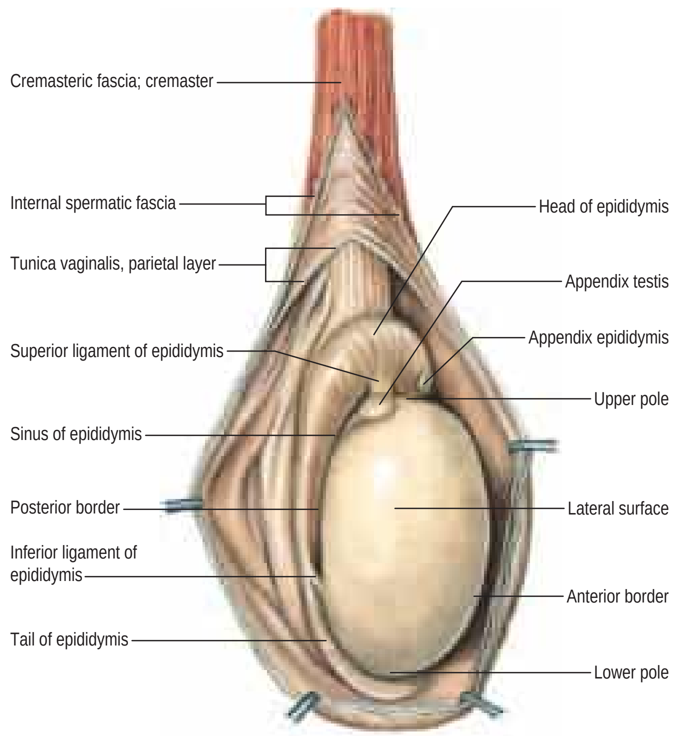

Fig. 76.4 The left testis, exposed by incising and laying open the cremasteric fascia and parietal layer of the tunica vaginalis on the lateral aspect of the testis. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban and Fischer. Copyright 2013.)

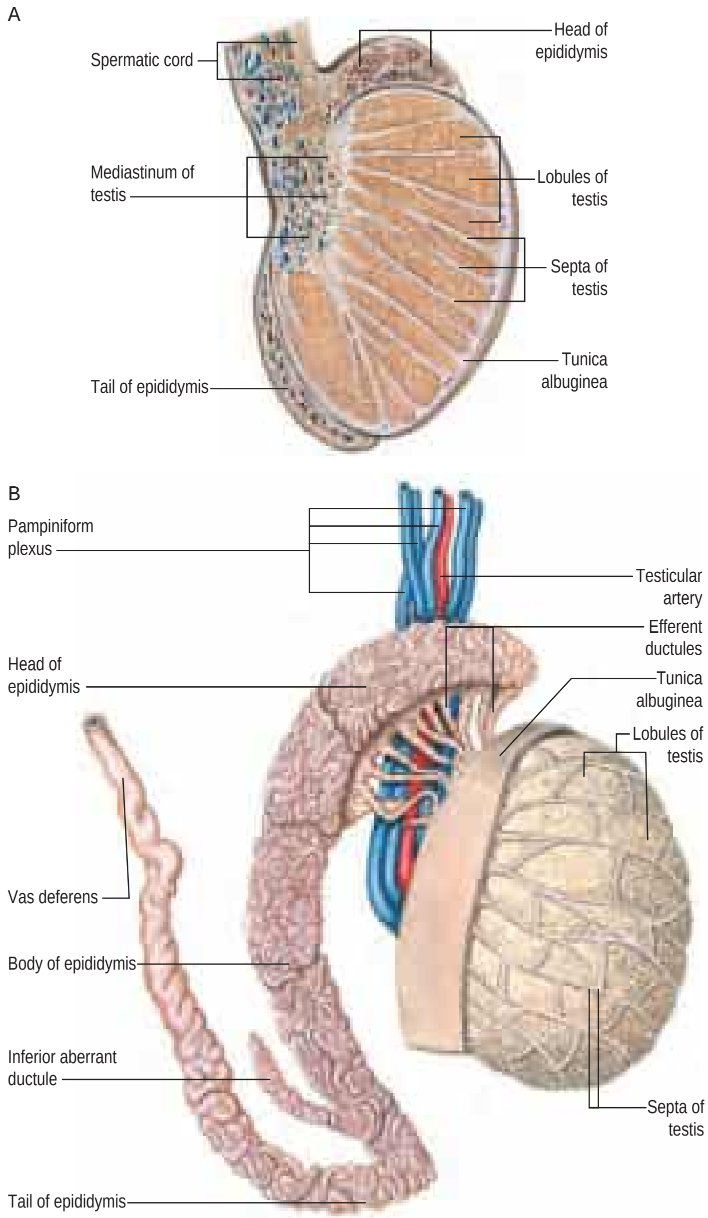

Fig. 76.5 **A**, A vertical section through the testis and epididymis. **B**, The arrangement of the ducts of the testis and the mode of formation of the vas deferens. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban and Fischer. Copyright 2013.)

of the renal veins, while the left testicular vein joins the left renal vein at a right angle (Fig. 76.8). The testicular veins contain valves.

Lymphatic drainage

Testicular lymphatic flow is abundant and consistent, and follows the general retroperitoneal scheme of vertical drainage, with lateral flow from right to left. Lymphatic vessels from the right testis drain primarily into the inter-aortocaval nodes and paracaval nodes, with some drainage into the left para-aortic nodes. Lymphatic vessels from the left testis drain into the left para-aortic and inter-aortocaval nodes.

Innervation

The testis is innervated either by nerve fibres that arise from the tenth and eleventh thoracic spinal segments, via the renal and aortic plexuses, and accompany the testicular vessels, or by fibres that arise from the pelvic plexus and accompany the vas deferens (Rauchenwald et al 1995). Interestingly, some afferent and efferent nerves have been shown to cross over to the contralateral pelvic plexus (Taguchi et al 1999), which may be one reason why pathological processes in one testis can affect the other.

Microstructure

The testis is enclosed within a tough, collagenous tunica albuginea, which thickens posteriorly as the mediastinum testis. Blood vessels, lymphatics and genital ducts all enter and leave the testis at the mediastinum (Fig. 76.9). Septations originating at the mediastinum testis extend internally to partition the testis into approximately 250 lobules that differ in size, the largest and longest lobules being near the centre (see Fig. 76.5). Each lobule contains 1–4 convoluted seminiferous tubules and interstitial tissue composed of Leydig cells, mast cells, macrophages, nerves and blood vessels.

Seminiferous tubules are typically long, highly coiled and looped; both ends terminate in the mediastinum testis. Spermatogenesis occurs in the highly coiled portions (Fig. 76.10). As the tubules reach the apical portion of the lobule near the mediastinum, they become much less convoluted and form short tubuli recti, which lack spermatogenic cells and are, instead, lined by cuboidal epithelium. Within the mediastinum testis, tubuli recti anastomose to form the rete testis, which is lined by a flat epithelium. Here, tubular fluid is reabsorbed and spermatozoa become concentrated. The rete testis coalesces to form 7–15 efferent ductules, which act as conduits to carry spermatozoa into the caput epididymis. The efferent ductules are lined by a ciliated columnar epithelium that also contains non-ciliated, actively endocytic cells. Outside the epithelial lining, the ductules are surrounded by a thin circular coat of smooth muscle.

There is a reduction in the diameter of the seminiferous tubules during the third trimester of pregnancy and immediately before birth, which is followed by a gradual increase in the diameter throughout childhood (Mendez and Emery 1979). Seminiferous tubules are responsible for up to 80% of the total testicular volume. It is estimated that the combined length of the 600–1200 tubules in the human testis is approximately 250 metres. Each tubule is surrounded by a basal lamina, resting on a complex, stratified epithelium. In cross-section, the lumen of the seminiferous tubule is lined by Sertoli cells, and contains spermatozoa in various stages of development, from spermatogonial stem cells near the base, to progressively mature forms (spermatogonia, spermatocytes, spermatids and spermatozoa) arranged in a systematic fashion towards the centre of the tubule. Residual bodies, spherical structures derived from surplus spermatid cytoplasm shed during spermatozoal maturation, may be found among the spermatids.

Specialized tight junctional complexes between adjacent Sertoli cells form a functional 'blood–testis barrier' that subdivides the seminiferous epithelium into basal and adluminal compartments. Spermatogonia and developing spermatocytes lie outside the blood–testis barrier, in the basal compartment, whereas mature spermatocytes and spermatids are sequestered above the blood-testis barrier, in the adluminal compartment.

The longer length of the left testicular vein and its angle of insertion into the left renal vein have been suggested as possible aetiologies for the occurrence of varicoceles. Additionally, the testicular veins may anastomose with the external pudendal, cremasteric and vasal veins, allowing varicoceles to persist or recur after ablative procedures that do not address these potential collaterals.

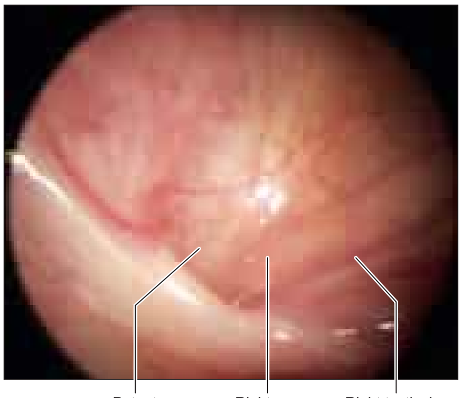

Fig. 76.6 A laparoscopic view of a persistent communication between the scrotum and peritoneal cavity in a 4-month-old boy (patent processus vaginalis).

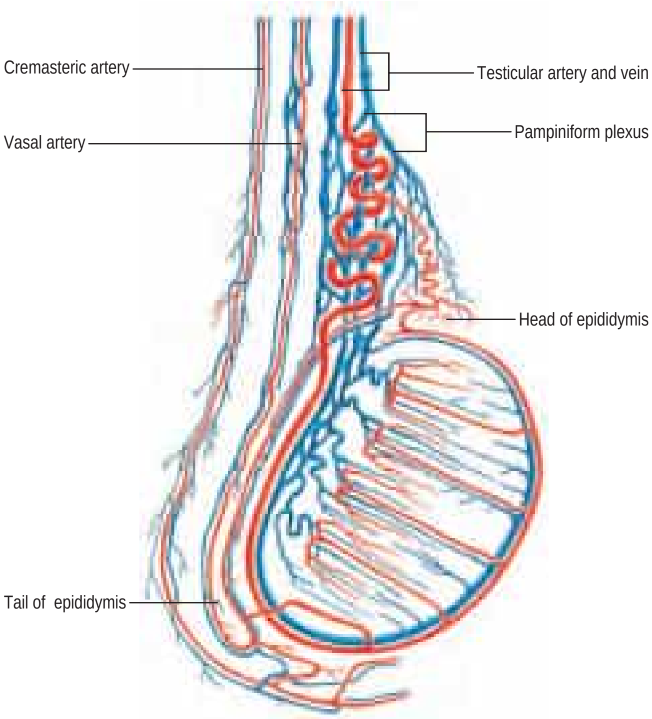

Fig. 76.7 The arterial blood supply and venous drainage of the testis. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban and Fischer. Copyright 2013.)

Fig. 76.8 A multislice computed tomogram of the inferior vena cava, showing the left testicular vein draining to the left renal vein, and the right testicular vein draining directly to the inferior vena cava. Key: 1, right testicular vein; 2, inferior vena cava; 3, left renal vein; 4, left testicular vein.

Fig. 76.9 A colour Doppler scan of the scrotal contents showing normal flow. The linear echogenic band (arrow) seen centrally represents the mediastinum testis, which is composed of fibrofatty material.

Spermatogonia Spermatogonia, the germ cells for all spermatozoa, are descended from primordial germ cells that migrate into the genital cords of the developing testis. In the fully differentiated testis, they are located along the basal laminae of the seminiferous tubules (Fig. 76.10B). Based on their cellular and nuclear dimensions, distribution of nuclear chromatin, and histochemical and ultrastructural properties, spermatogonia are characterized as either dark-type A (Ad), pale-type A (Ap) or type B. Ad spermatogonia divide mitotically to maintain their own population, and can differentiate to give rise to Ap cells, the precursors of type B cells, which are committed to the spermatogenesis cycle. Type B cells give rise to type I spermatocytes; they leave the basal compartment of the seminiferous tubule and cross the blood–testis barrier to enter the adluminal compartment in a step coordinated by Sertoli cells.

The generation of mature spermatozoa from spermatogonia takes approximately 64 days. In cross-section, a seminiferous tubule shows more than one phase of the cycle around its circumference because waves of progression through a spermatogenic cycle occur in spirals along the length of a tubule (see Fig. 76.10B).

Primary and secondary spermatocytes Primary spermatocytes have a diploid chromosome number, with duplicated sister chromatids (4N DNA content). They are large cells with round nuclei, in which the chromatin is condensed into dark, thread-like, coiled chromatids. They undergo meiosis I to give rise to secondary spermatids (2N), which rapidly undergo meiosis II to form haploid spermatids (N). Theoretically, each primary spermatocyte produces four spermatids; however, a proportion of spermatids degenerate during maturation, reducing the expected yield.

Spermatids Spermatids undergo a series of nuclear and cytoplasmic changes, termed spermiogenesis, in order to develop into mature spermatozoa. These changes take place while the spermatids are closely associated with Sertoli cells, and linked to one another by cytoplasmic bridges. Spermiogenesis includes the development of the acrosome

CHAPTER 76

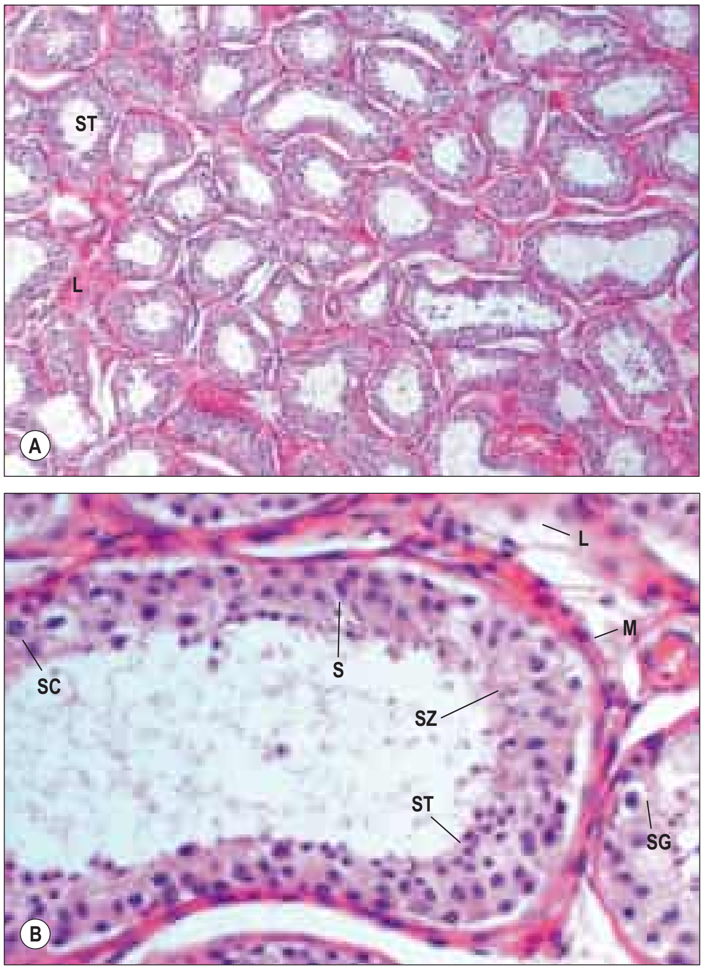

Fig. 76.10 **A**, Seminiferous tubules (ST; cut in various planes of section) and the interstitial tissue (Leydig cells, L) of the testis. The seminiferous tubules are highly convoluted and lined by a stratified epithelium, which consists of cells in various stages of spermatogenesis and spermiogenesis (collectively referred to as the spermatogenic series). Non-spermatogenic cells are the Sertoli cells. **B**, A human seminiferous tubule showing the differentiation sequence of spermatozoa from basally situated spermatogonia (SG). Large primary spermatocytes (SC) have characteristic thread-like chromatin in various stages of prophase of the first meiotic division. Smaller haploid spermatids (ST) have round nuclei initially, but mature to possess the dense, elongated nuclei and flagella of spermatozoa (SZ). Sertoli cells (S) are identified from their oval or pear-shaped nuclei, orientated perpendicular to the basal lamina, and from their prominent nucleoli. The tubule is surrounded by peritubular myoid cells (M). Clusters of large endocrine Leydig cells (L) are seen in the interstitial connective tissue.

from Golgi vesicles, generation of the axoneme from the centrioles, and formation of the acrosomal cap at the anterior pole of the spermatozoon. Concurrently, the nuclear chromatin condenses and the nucleus assumes a spearhead shape. The cytoplasmic volume shrinks, bringing the wall of the acrosomal vesicle into contact with the plasma membrane. A perinuclear sheath of microtubules develops from the posterior edge of the acrosome and extends towards the posterior pole of the spermatozoon. Here, the microtubules are arranged in the typical flagellar pattern of nine outer microtubules surrounding two central microtubules, the axonemal complex that extends longitudinally to form the tail of the spermatozoon. Mitochondria migrate along the axonemal complex and concentrate in the mid-piece of the tail. In the final phase of spermiogenesis, excess cytoplasm is detached from the spermatozoon as a residual body that is phagocytosed and degraded by Sertoli cells. During the formation of residual bodies, spermatids lose their cytoplasmic bridges, and separate from one another, before being released into the lumen of the seminiferous tubule.

Spermatozoa Spermatozoa released into the seminiferous tubule are structurally mature but usually non-motile. In the presence of excurrent ductal obstruction, testicular sperm can acquire motility (Jow et al 1993). The shape of the spermatozoon is ideally suited for rapid progressive motility (Fig. 76.11). Its head has minimal cytoplasm and measures approximately 4 μm×3 μm. It contains an elongated, flattened nucleus with condensed, dark-staining chromatin, covered anteriorly by an acrosomal cap. The latter contains acid phosphatase, hyaluronidase and proteases necessary for oocyte fertilization. The head is connected by a short neck, approximately 0.3 μm in length, to a long tail, which is divided into middle, principal and end pieces. The mid-piece of the tail is a long cylinder, approximately 7 μm in length. It consists of an axial bundle of microtubules, the axoneme, surrounded by a cylinder of nine dense outer microtubules, surrounded by a helical mitochondrial sheath. The mid-piece is the powerhouse of the spermatozoon. The principal piece of the tail is responsible for motility. With a length of 40 μm and a diameter of 0.5 μm, the tail forms the majority of the volume of a spermatozoon. The axonemal complex is continuous from the neck region to the terminus of the tail, with only the axoneme persisting in the final 5–7 μm.

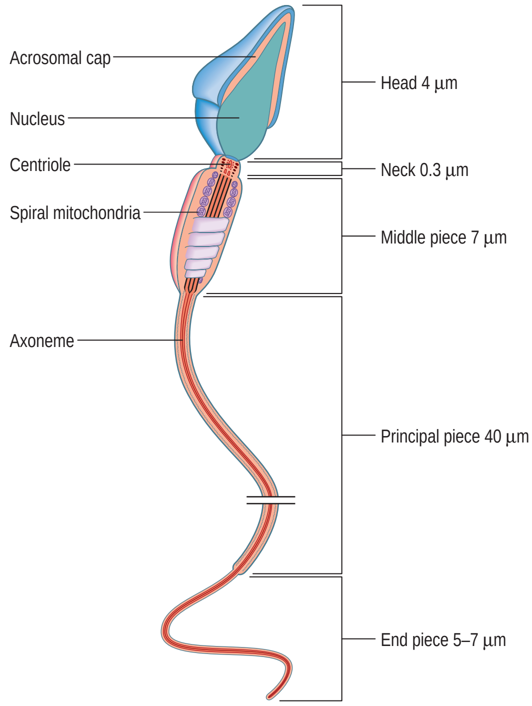

Fig. 76.11 The main ultrastructural features of a mature spermatozoon.

Sertoli cells Sertoli cells have euchromatic and irregular nuclei that are aligned perpendicular to the basal lamina and contain one or two prominent nucleoli. They are variable in shape. Their basal end rests on the basal lamina, while their apical end extends into the tubule lumen. Consistent with their phagocytic function, their cytoplasm is rich in lysosomes. Sertoli cells provide key support for spermatogenesis within the seminiferous tubules. Complex recesses in their plasma membranes serve to envelop spermatogonia, spermatids and spermatozoa, until the latter are mature enough for release. Long cytoplasmic processes extend between spermatogonia in the basal compartment and spermatocytes in the adluminal compartment of the seminiferous tubule. Adjacent Sertoli cells are joined at this level by tight junctions that create a diffusion barrier between the extratubular and intratubular compartments. This so-called blood–testis barrier can be disrupted by traumatic or inflammatory events, allowing self-directed immune responses to develop against sperm antigens, potentially leading to subfertility.

In addition to their role in spermatogenesis, Sertoli cells secrete proteinaceous fluid to facilitate spermatozoal transport through the seminiferous tubules and into the excurrent ducts. They also regulate the intratesticular hormonal milieu by secreting inhibin B and androgen binding protein in response to stimulation by pituitary gonadotropins.

Leydig cells and interstitial tissue The interstitial tissue between seminiferous tubules includes peritubular myoid cells, vessels, nerves and clusters of Leydig cells. Myoid cells are contractile; their rhythmic activity helps propel non-motile spermatozoa through the seminiferous tubules towards the rete testis and excurrent ductal system. Leydig cells are large polyhedral cells with eccentric nuclei containing 1–3 nucleoli, and pale-staining cytoplasm containing smooth endoplasmic reticulum, lipid droplets, and characteristic needle-shaped, crystalloid inclusions (crystals of Reinke). In response to stimulation by pituitary gonadotropins, they synthesize and secrete testosterone.

Age-related changes Functionally, the fetal testis is primarily an endocrine organ that produces testosterone and anti-Müllerian hormone. Seminiferous tubules do not become canalized until approximately the seventh month of gestation. Fetal Leydig cells, responsible for androgen-induced differentiation of male genitalia, degenerate after birth. A second wave of Leydig cell differentiation occurs 2–3 months after birth, briefly elevating testosterone levels in male infants. The Leydig cells of early infancy subsequently regress and the testis remains in a state of dormancy during childhood. In a study of Japanese boys, gonocytes were reported to appear at about the age of 2 years, spermatocytes by 4 years and spermatids by 11 years (Yuasa et al 2001).

Puberty is associated with the development of an adult population of androgen-producing Leydig cells, which persist throughout adult life. The testes grow slowly in size until the age of 10 or 11 years, at which time there is a marked acceleration of growth rate. This increase in testicular size is largely due to the onset of spermatogenesis, characterized by proliferation and differentiation of previously dormant spermatogonial stem cells. Testicular size, sperm quality and quantity, and the numbers of Sertoli and Leydig cells have all been reported to decline with age, although no consistent or definitive age for the onset of this decline has been identified. Testicular volume occupied by seminiferous tubules decreases, whereas that occupied by interstitial tissue remains approximately constant.

The most frequently observed histological change in the ageing testis is the apparent variability in spermatogenesis; spermatogenesis is complete in some areas, and reduced in others, or absent altogether as a result of tubular sclerosis. In tubules where spermatogenesis is complete, morphological abnormalities, such as multinucleation, may be observed in the germ cells. Germ cell loss, beginning with spermatids, and progressively affecting the earlier stages of spermatogenesis, can also be seen. As a result, both sperm quality and quantity may be affected. In some men, this change is notable as early as the third or fourth decade of life.

Sertoli cells are also affected by ageing, and show a range of morphological changes including de-differentiation, mitochondrial metaplasia, and multinucleation. In Leydig cells, there is a decline in the number of mitochondria and the quantity of smooth endoplasmic reticulum, paralleled by an increase in lipid droplets, crystalline inclusions and residual bodies. Some cells may also become multinucleated. Functionally, these changes are manifested as an age-related gradual decline in circulating testosterone levels.

Sperm motility and maturation Human spermatozoa acquire an increased capacity for motility as they migrate through the epididymis, which is manifested not only as a quantitative increase in the percentage of spermatozoa with motile tails, but also as a qualitative change from an immature to a more mature pattern of motility. Spermatozoa in the proximal epididymis demonstrate high-amplitude, low-frequency tail movements, producing little motion. This is in contrast to spermatozoa in the cauda epididymis, which demonstrate low-amplitude and high-frequency tail movements, resulting in considerably greater forward progression (Bedford et al 1973).

Whether, and to what extent, sperm motility is dependent on the interaction of human spermatozoa with a particular section of the epididymis is unknown. In patients with congenital absence or obstruction of the vas deferens, spermatozoa in the distal epididymis demonstrate worse motility compared to spermatozoa in the proximal epididymis (Silber 1989, Schoysman and Bedford 1986). Therefore, intrinsic sperm processes, as well as intrinsic epididymal function, appear to play an important role in sperm maturation.

Following ejaculation, spermatozoa display their full pattern of motility. Although sperm motility is highest in the hours following ejaculation, motile human spermatozoa have been recovered from cervical mucus several days after insemination. However, this survival period may be of little relevance, given that human spermatozoa are capable of migrating to their tubal destination within an hour of insemination.

After entering the female reproductive tract, spermatozoa undergo a process known as capacitation, to render them capable of fertilizing the oocyte. Capacitation involves a number of structural and biochemical changes, including the development of hyperactivated motility and completion of the acrosome reaction. Independent of other sperm characteristics such as motility and morphology, capacitation is a necessary step for the development of functional spermatozoa.

Developmental anomalies of the testis

Cryptorchidism Testicular descent from the abdominal cavity to the scrotum may be arrested at any point along its course at the deep inguinal ring, in the inguinal canal, or between the superficial inguinal ring and the scrotum. Retention in the inguinal canal is associated with a patent processus vaginalis, and may be further complicated by a congenital hernia. Occasionally, the testis may migrate outside its normal path of descent, and lie in an ectopic location.

At birth, 3% of full-term male infants have a unilateral undescended testis. By 6 months of age, this number decreases to less than 1%. Undescended testes are associated with a higher risk of infertility and testicular cancer later in life.

In contrast, maldescent in children between the ages of 1 and 15 years is associated with Sertoli cell degeneration (Rune et al 1992).

Retractile testis

Available with the *Gray's Anatomy* e-book

Torsion

Available with the *Gray's Anatomy* e-book

Hydrocele

Available with the *Gray's Anatomy* e-book

Splenogonadal fusion

Available with the *Gray's Anatomy* e-book

Transverse/crossed testicular ectopia

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Polyorchidism

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EPIDIDYMIS

The epididymis lies posterior, and slightly lateral, to the testis. Anatomically, it is divided into three sections: caput (head), corpus (body) and cauda (tail). Between 8 and 12 efferent ductules from the superior pole of the testis drain into and form the caput epididymis. Distally, the cauda epididymis becomes continuous with the convoluted portion of the vas deferens (see Fig. 76.5B). The epididymis is invested by tunica vaginalis, continuous with that covering the testis. Laterally, a deep groove, the sinus epididymis, marks the boundary between the testis and epididymis. Testis–epididymis non-fusion has been described in children with cryptorchidism; non-fusion may involve the epididymal head or tail, or the whole epididymis (Kraft et al 2011).

Epididymal cyst and spermatoceles

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Microstructure

The epididymal tubule is 3–4 metres in length (Turner et al 1978), and is entirely encapsulated by the tunica vaginalis. Extensions from this connective tissue sheath enter the interductal spaces, forming septa that divide the tubule into histologically similar regions (Kormano and Reijonen 1976). The caput epididymis consists of 8–12 efferent ducts and the proximal segment of the ductus epididymis. The efferent ducts become larger and more convoluted than they are in the testis. Each duct is 15–20 cm in length, and opens into an individual epididymal tubule. The tubules anastomose with one another along the length of the epididymis, eventually becoming a single, coiled tubule in the corpus epididymis. They are surrounded by contractile smooth muscle cells (**Fig. 76.13**), which increase in size and number in the distal epididymis. Peristaltic contractions generated by the smooth muscle propel the spermatozoa in an antegrade direction towards the cauda.

Epithelium The epithelial lining of the epididymal tubule contains principal, basal, apical and clear cells. Principal cells are tall columnar cells with basally located, oval nuclei. They bear long stereocilia (15 µm), and function to resorb testicular fluid; approximately 90% of

Orchidopexy is, therefore, recommended for patients with persistent cryptorchidism after 6 months of age and within 18 months of age. Available evidence suggests that orchidopexy at any age may improve spermatogenesis (Shin et al 1997), but does not decrease the risk of testicular cancer in a cryptorchid testis. An inverse relationship has been reported between age at orchidopexy and total sperm count and sperm motility. The data are in support of orchidopexy in the first year of life (Canavese et al 2009). Nevertheless, surgical correction does maximize the chance of early detection of a testicular mass, and is the only means of restoring a normal milieu for spermatogenesis to occur. Interestingly, Leydig cell function is usually unchanged by maldescent, so serum testosterone levels remain within the normal range in affected patients.

Retractile testes are seen more commonly than undescended testes in boys. A retractile testis is one that moves to and fro between the groin and the scrotum. Unlike undescended testes, retractile testes can be manipulated to the lower part of the scrotum on clinical examination. Retractile testes are smaller than normal testes, having a mean volume of 0.50 ml in children.

Testicular torsion refers to rotation of the testis around its blood supply, leading to testicular ischaemia. Torsion may be extravaginal or intravaginal, depending on whether it involves a rotation of both the testis and the tunica vaginalis, or of the testis alone, within an intact tunica vaginalis. In either case, torsion results in severe scrotal pain secondary to tissue ischaemia. Fertility can be affected by a single episode of torsion. If unrelieved within 4–6 hours, permanent tissue loss can occur. Testicular torsion is therefore considered a surgical emergency.

Other structures in the scrotum, such as the appendix testis (hydatid of Morgagni) (**Fig. 76.12**) and appendix epididymis (see Fig. 76.4), can also undergo torsion, resulting in scrotal pain that may be difficult to differentiate from testicular torsion. In some instances, a 'blue dot' sign is noted at the upper pole of the testis, which is diagnostic. These structures are developmental remnants of the paramesonephric (Müllerian) duct and the mesonephros, respectively. There are no associated long-term sequelae.

A patent processus vaginalis allows communication between the peritoneal cavity and the spermatic cord or scrotum. Passage of peritoneal fluid into the scrotum presents as a communicating hydrocele, and usually resolves spontaneously once the processus is obliterated, by 18–24 months of age. Alternatively, if the processus is patent proximally but obliterated distally, a hydrocele or cyst of the cord may be noted. Surgical treatment may be indicated for persistent, non-communicating hydroceles. Hydrocele can also occur as a result of lymphatic obstruction from testicular tumour, epididymitis, orchitis or trauma. A 20% incidence of hydrocele has been reported in children having laparoscopic *en bloc* ligation of the spermatic vessels on the posterior abdominal wall (Palomo procedure). This complication is avoided by lymphatic-sparing surgery (Schwentner et al 2006).

Splenogonadal fusion is a rare congenital anomaly characterized by splenic tissue being connected to the gonad, resulting from an abnormal connection between the spleen and the gonad during gestation. It has a higher predominance in males and is almost invariably on the left side. It may be of the continuous type, where the testis is connected to the main spleen, or of the discontinuous type, where the testis is connected to ectopic splenic tissue.

Fig. 76.12 Gangrene of the appendix testis secondary to torsion in an 8-year-old child.

Transverse/crossed testicular ectopia is an extremely rare condition in which both testes descend into a single inguinal canal or hemiscrotum. The origin of the spermatic cord is normally located on each side. It may occur in association with persistent Müllerian duct syndrome (Tiryaki et al 2005).

Polyorchidism is a rare congenital anomaly in which there are two or more testes. The most common type consists of three testes; cases with four testes have also been reported. The supernumerary testes may or may not be connected to a vas deferens.

Epididymal cysts arise from the epididymal tubules and may occur anywhere in the caput, corpus or cauda epididymis. If asymptomatic, removal is unnecessary, particularly given the risk of iatrogenic epididymal obstruction. Spermatoceles are cysts that are, basically, aneurysms of the efferent ducts that form the caput epididymis and contain sperm. Removal is unnecessary unless they grow to a large size and cause pain (Kaufman et al 2011). However, spermatoceles may be aspirated as a source of spermatozoa.

the total secretory fluid volume is absorbed in the epididymis. They also endocytose other components of seminal fluid and produce glycoproteins that are essential for sperm maturation. Basal cells lie between the bases of the principal cells and are thought to be precursors of principal cells. Apical and clear cells are far less common than principal and basal cells. Apical cells are rich in mitochondria and are most abundant in the caput epididymis. In contrast, clear cells are columnar and most abundant in the cauda epididymis. They have few microvilli but numerous endocytic vesicles and lipid droplets. Their functions are unknown.

VAS DEFERENS, SPERMATIC CORD, PARADIDYMIS AND EJACULATORY DUCT

VAS DEFERENS

The vas deferens (ductus deferens) is a tubular structure derived from the mesonephric duct. Its primary function is the transport of sperm from the epididymis to the urethra, although absorptive and secretory functions have also been described (Hoffer 1976). As it arises from the cauda epididymis, the vas is tortuous for 2–3 cm (see Fig. 76.5B). Beyond this convoluted segment, it lies posterior and parallel to the vessels of the spermatic cord, passes through the inguinal canal (see Fig. 76.16), and emerges in the pelvis lateral to the inferior epigastric vessels (see Fig. 61.3). At the internal ring, the vas diverges from the testicular vessels, coursing medial to the structures of the pelvic side wall, in order to reach the base of the prostate posteriorly. Here, the vas once again becomes tortuous and dilated (the ampulla of the vas deferens), before culminating at the ejaculatory duct (see Fig. 74.23).

Vascular supply and lymphatic drainage

The vas deferens is supplied by the vasal artery, which is usually derived from the superior vesical artery, and occasionally from the inferior vesical artery, both branches of the internal iliac artery. Rarely, it is derived directly from the internal iliac artery. Veins from the vas and seminal vesicles drain to the pelvic venous plexus, whereas the associated lymphatic vessels drain into the external and internal iliac nodes.

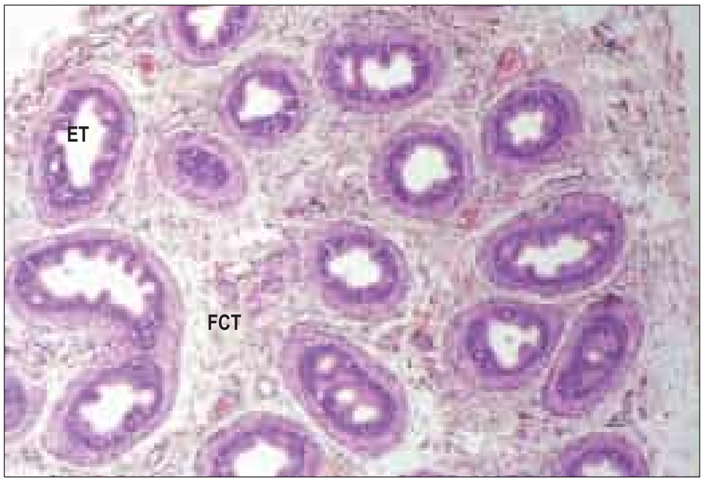

Fig. 76.13 The microstructure of the epididymis. Abbreviations: ET, epididymal tubule; FCT, fibromuscular connective tissue.

Innervation

The vasa deferentia are innervated by a rich autonomic plexus of primarily postganglionic sympathetic fibres derived from the pelvic plexus.

Microstructure

In humans, the vas deferens is 30–35 cm in length and 2–3 mm in diameter, with a luminal diameter of 300–500 µm. In cross-section, it consists of an outer adventitial sheath containing blood vessels and nerves; a thick, three-layered muscular wall of inner and outer longitudinal and middle circular smooth muscle; and an inner mucosal lining of pseudostratified columnar epithelium with non-motile cilia (**Fig. 76.14**). It has the greatest muscle to lumen ratio (approximately 10:1) of any hollow viscus in the body. During ejaculation, the smooth muscle layers contract reflexively, propelling the sperm in an antegrade direction (see above).

Developmental anomalies of the vas deferens

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SPERMATIC CORD

The spermatic cord begins at the deep or internal inguinal ring, extends the length of the inguinal canal, exits the canal at the superficial or external inguinal ring, and suspends the testis in the scrotum (**Fig. 76.15**; see Fig. 76.3). Between the superficial inguinal ring and the testis, the cord lies anterior to the tendon of adductor longus. It is flanked anteriorly by the superficial external pudendal artery, and posteriorly by the deep external pudendal artery. The ilioinguinal nerve lies inferior to the cord as it traverses the inguinal canal.

In addition to the artery, veins, lymphatics and nerves supplying the testis, which are contained within the internal spermatic fascia, the spermatic cord includes the ilioinguinal nerve, the genital branch of the genitofemoral nerve, the cremasteric artery, veins and lymphatics, and the vasal artery, veins and lymphatics, all of which are contained within the external spermatic fascia (**Fig. 76.16**; see Fig. 76.2). These fascial layers are continuous with the layers of the abdominal wall. The internal spermatic fascia is derived from the transversalis fascia, and forms a thin, loose layer around the spermatic cord. The cremasteric fascia, which contains the skeletal muscle fibres that make up the cremaster muscle, is derived from internal oblique. The external spermatic fascia is continuous with the aponeurosis of external oblique.

Ectopic suprarenal tissue (adrenal rest) is encountered within the distal end of the spermatic cord in 2% of children undergoing inguinal procedures. Adrenal rests are typically bright yellow to orange in colour, round or oval in shape, and up to 5 mm in diameter; they contain the three layers of the suprarenal (adrenal cortex) but no medulla (Savas et al 2001).

Fig. 76.14 The vas deferens. **A**, A low-power view, transverse section. **B**, A higher-power view. The convoluted lumen is lined by pseudostratified columnar epithelial cells.

The vas deferens may be congenitally absent, either unilaterally or bilaterally. When bilateral, the condition is associated with a mutation of the cystic fibrosis transmembrane conductance gene in 80% of affected men. This condition is characterized by low semen volume due to absent or hypoplastic seminal vesicles and azoospermia, although spermatogenesis is usually normal. Absent vasa is usually the only genital manifestation of cystic fibrosis.

The anatomy of the vas deferens is worthy of note in children with cryptorchidism (see above). At laparoscopic orchidopexy for intra-abdominal testis, it is important to exclude a 'looping vas deferens' that enters the inguinal canal and loops back to the abdominal cavity (Shalaby et al 2011); failure to recognize this anatomical anomaly may result in iatrogenic injury to the vas deferens.

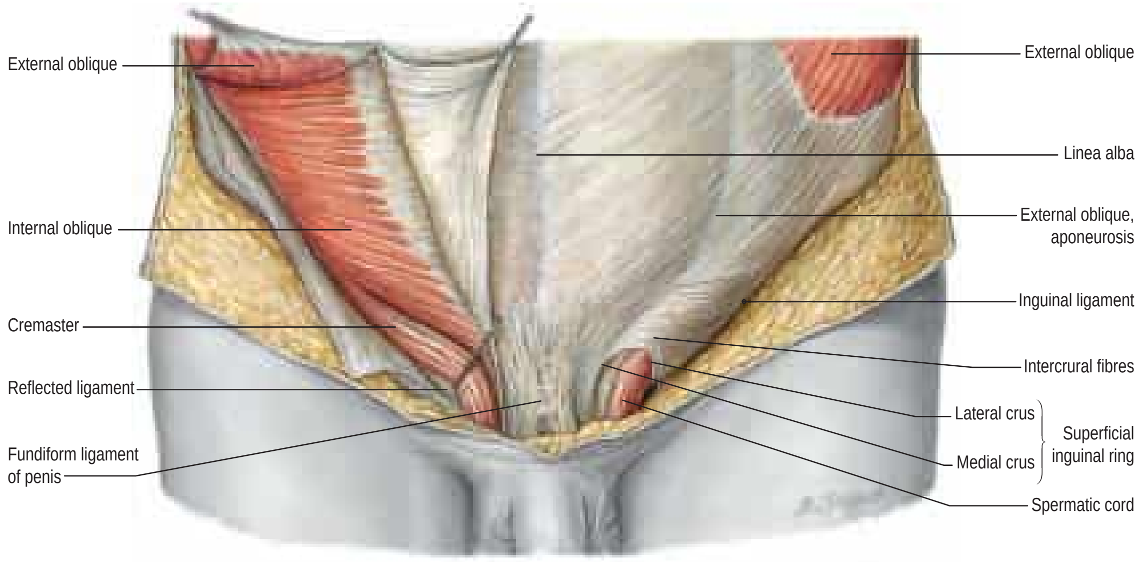

Fig. 76.15 The relationship of the spermatic cord to the anterior abdominal wall. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban and Fischer. Copyright 2013.)

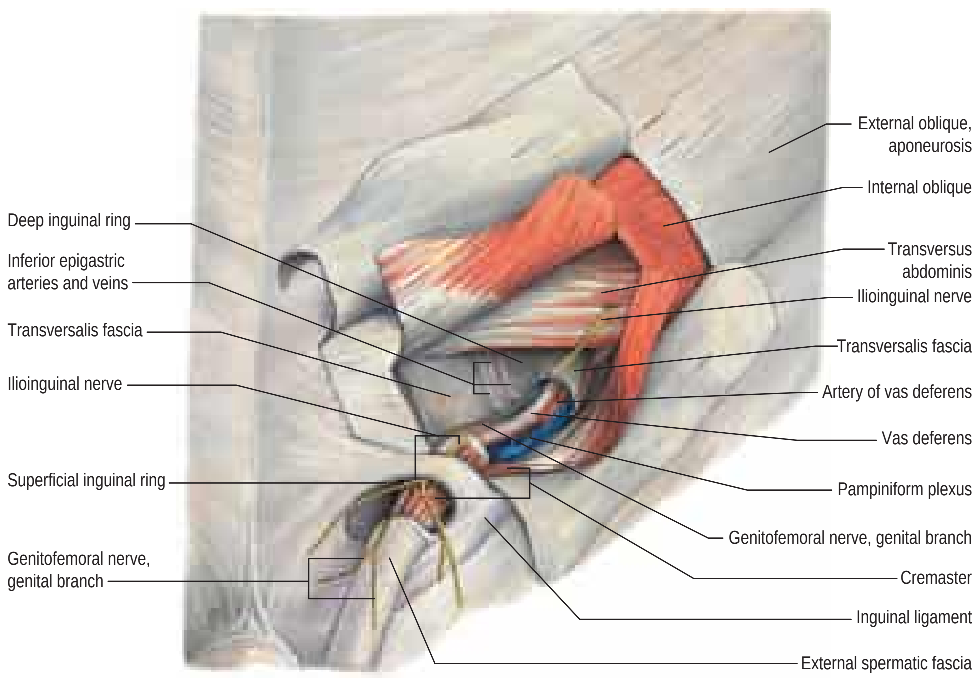

Fig. 76.16 Structures contained within the spermatic cord. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban and Fischer. Copyright 2013.)

PARADIDYMIS

The paradidymis (organ of Giraldes) is a small collection of convoluted tubules located anteriorly in the spermatic cord, just above the caput epididymis. The tubules are lined with columnar ciliated epithelium and are likely to represent a vestigial remnant of the mesonephros.

EJACULATORY DUCTS

The paired ejaculatory ducts are formed from the union of the duct of the seminal vesicle with the ampulla of the vas deferens. Each is approximately 2 cm in length, and extends from the base of the prostate, between the median and lateral lobes, towards its opening on the verumontanum (see Fig. 75.13). The ducts diminish in size and converge towards their ends. Ejaculatory duct obstruction, though rare, can lead to oligospermia or azoospermia. Causes include congenital cysts, such as Müllerian duct cysts, or iatrogenic injury during urethral manipulation. In contrast to the walls of the vas deferens, the walls of the ejaculatory ducts are thin. They consist of an outer fibrous layer, which decreases in thickness on their entry into the prostate; a thin layer of smooth muscle fibres; and a mucosa lined by columnar epithelium. The ducts dilate during ejaculation. Normal luminal and wall dimensions of the ejaculatory duct are remarkably uniform among men; a luminal diameter of greater than 2.3 mm defines a dilated or obstructed efferent duct (Nguyen et al 1996).

ACCESSORY GLANDULAR STRUCTURES

SEMINAL VESICLES

The seminal vesicles are paired outpouchings of the terminal vas deferens, located at the base of the prostate, between the bladder and the rectum (**Fig. 76.17**, see Fig. 74.23). In adults, the seminal vesicle measures between 5 and 10 cm in length, and 3 and 5 cm in diameter, with an average volume capacity of 13 ml (Goldstein 2012). In the majority of men, the right seminal vesicle is slightly larger than the left; the size of both glands decreases with age.

In essence, each seminal vesicle is a coiled tube with irregular diverticula, contained within a dense, fibromuscular sheath and partly covered by peritoneum. The upper pole is a cul-de-sac, while the lower pole narrows to a straight duct, which, together with the vas deferens, culminates as the ejaculatory duct. The seminal vesicles are intimately associated with the prostate and bladder anteriorly, the distal ureter superiorly, and the rectum and Denonvilliers' fascia posteriorly. The ampullae of the vas deferens lie along the medial margins of the seminal vesicles, while the veins of the prostatic venous plexus lie laterally.

Fig. 76.17 An axial magnetic resonance imaging (MRI) scan demonstrating the normal high signal in the seminal vesicles on a T2-weighted scan.

Agenesis of the seminal vesicles is a congenital anomaly, which is associated with absence of the vas deferens or vasoureteral communication (Wu et al 2005).

Vascular supply and lymphatic drainage

The primary blood supply to the seminal vesicles is from the vesiculodeferential artery, a branch of the umbilical artery (Braithwaite 1952). An additional source of blood supply is the inferior vesical artery, which arises from the internal iliac artery or the inferior gluteal artery. Venous drainage is provided by the vesiculodeferential veins and the inferior vesical plexus. Lymphatic drainage, accordingly, occurs via the internal iliac nodes.

Innervation

The seminal vesicles receive preganglionic parasympathetic input from the pelvic nerve, and sympathetic (postganglionic) and parasympathetic (preganglionic) input from the hypogastric nerve.

Microstructure

After puberty, the vesicles mature into elongated, sac-like structures, produce a viscous white–yellow fluid that contributes to at least two-thirds of the total ejaculate volume, and play an important role in sperm motility and metabolism. The secreted fluid is rich in fructose, coagulation proteins and prostaglandins, with a pH in the neutral to alkaline range. The wall of the seminal vesicle is composed of an external connective tissue layer, a middle smooth muscle layer (significantly thinner than the corresponding layer in the vas deferens), and an inner mucosal layer with a highly folded, labyrinthine structure (Fig. 76.18). The mucosal layer is lined by cuboidal to pseudostratified columnar epithelium, featuring typical protein-secreting cells. Contrary to the implication of their name, seminal vesicles are not reservoirs for spermatozoa, except in cases of ejaculatory duct obstruction; they contract during ejaculation to release secretions into the ejaculatory duct.

BULBOURETHRAL GLANDS

The bulbourethral glands (Cowper's glands) are small, round, yellow, lobulated structures, measuring approximately 1 cm in diameter, and

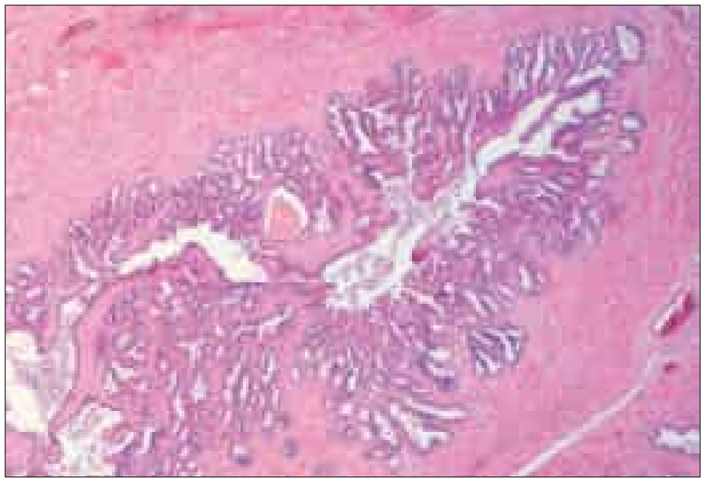

Fig. 76.18 The microstructure of the seminal vesicles, showing the tubulo-acinar structure.

located lateral to the membranous urethra and superior to the perineal membrane. They are drained by long excretory ducts, each duct being almost 3 cm in length, which pass obliquely and anteriorly from the region of the membranous urethra, to penetrate the perineal membrane, and open on the floor of the bulbar urethra, approximately 2.5 cm below the perineal membrane. The glands are surrounded by fibres of the urinary sphincter. During sexual excitement, contraction of the muscle fibres leads to expulsion of clear mucus from the glands into the bulbar urethra. Secretions of the bulbourethral glands comprise 5–10% of the total ejaculate volume.

Microstructure

Each bulbourethral gland consists of several lobules enclosed by a fibrous capsule. The secretory units are tubulo-alveolar in form. The glandular epithelium is columnar, and secretes acidic and neutral mucins into the urethra prior to ejaculation; the secretions primarily have a lubricating function. Diffuse mucosa-associated lymphoid tissue (MALT) is associated with the glands.

PERIURETHRAL GLANDS

The periurethral glands (glands of Littre) are most numerous in the penile urethra. Like the bulbourethral glands, they secrete mucus into the lumen of the urethra prior to ejaculation; the secretions have a lubricating function.

EXTERNAL GENITALIA

PENIS

The penis consists of an attached root in the perineum (radix), and a free, pendulous body (shaft), which is completely enveloped in skin (Fig. 76.19). At its base, the penis is supported by two suspensory ligaments, which anchor it to the pubic symphysis (Fig. 76.20B). These ligaments are composed primarily of elastic fibres and are continuous with Buck's fascia of the penis. The penile shaft contains three erectile columns – the paired corpora cavernosa and the corpus spongiosum; the urethra; and the investing fasciae, blood vessels and nerves associated with these structures (Fig. 76.20A).

The corpora cavernosa lie in intimate apposition with one another along the length of the penile shaft (Fig. 76.21); the corpus spongiosum lies in the ventral groove between the cavernous bodies. Proximally, posterior to the suspensory ligaments, the right and left corpora cavernosa diverge to form two tapering processes, the crura penis, which are firmly anchored to the ischiopubic rami (see Fig. 76.20B). The corpus spongiosum broadens between the two crura to form the bulbospongiosus muscle (see Fig. 76.20C). At the distal end of the penis, the corpus spongiosum again enlarges and assumes a bulbous shape to form the glans penis. The rounded base of the glans, the corona, separates the glans from the penile shaft. The glans is covered by the foreskin (prepuce), which is a loose fold of retractable skin attached to the ventral surface of the glans penis, under the corona, at the frenulum. Cutaneous sensitivity is greatest over the glans penis.

SECTION 8

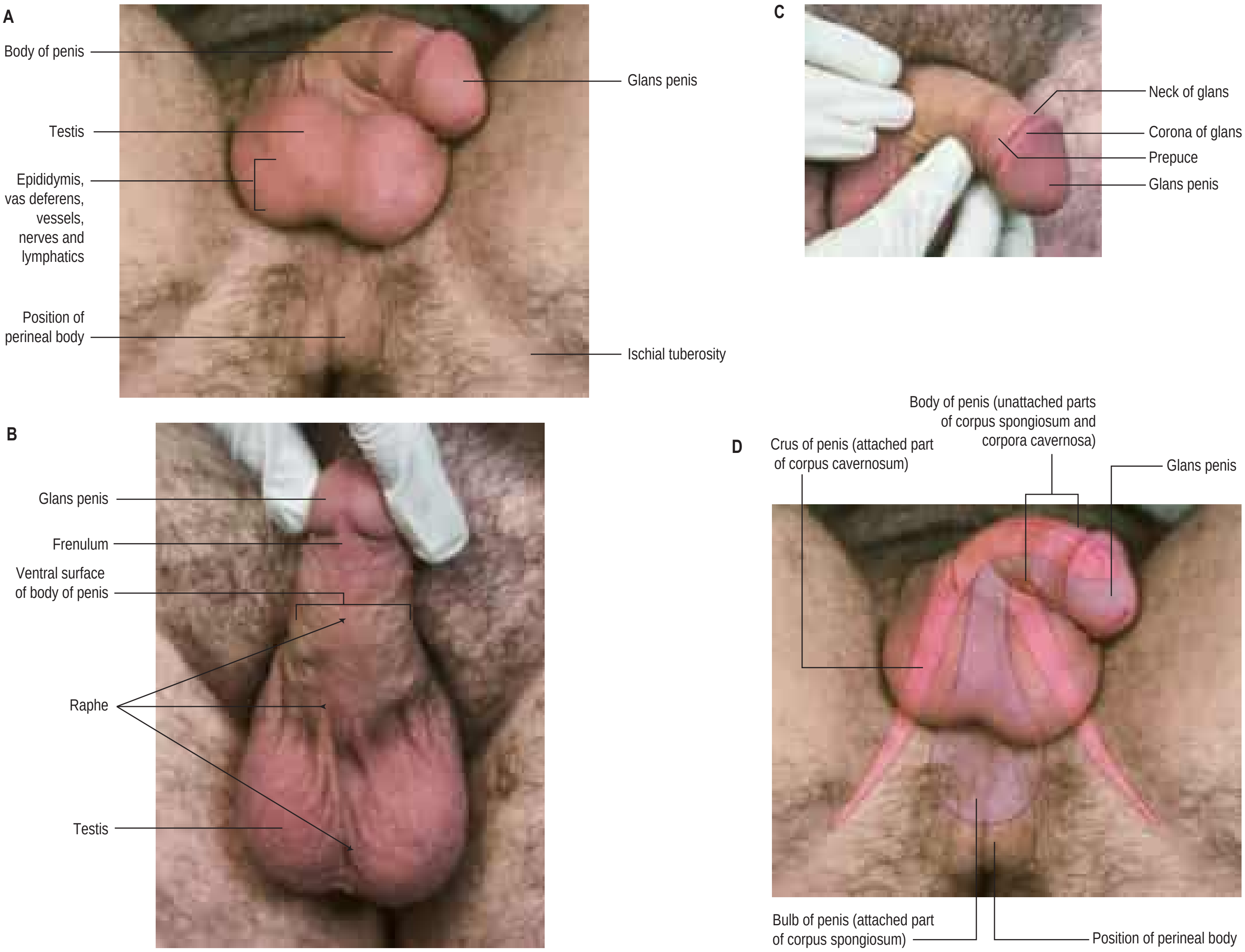

Fig. 76.19 Structures in the male urogenital triangle. **A**, An inferior view. **B**, The ventral surface of the body of the penis. **C**, A lateral view of the body of the penis and glans. **D**, An inferior view of the urogenital triangle of a male, with the erectile tissues of the penis indicated with overlays. (With permission from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

Skin The skin of the penile shaft is thin, highly elastic and devoid of appendages (hair or glandular elements), with the exception of smegma-producing glands located at the base of the corona. It is also devoid of fat and quite mobile because of loose attachments between the dartos fascia and the underlying Buck's fascia. In contrast, the skin of the glans is immobile, as a result of its direct attachment to the underlying tunica albuginea. Blood supply to the penile skin is independent of the erectile bodies, and is derived from the external pudendal branches of the femoral vessels that enter the base of the penis and run longitudinally within the dartos fascia, forming a rich anastomotic network. Thus, it is possible to mobilize penile shaft skin on a vascular pedicle for surgical procedures such as urethral reconstruction.

The superficial penile fascia is devoid of fat, and consists of loose connective tissue interspersed by fibres of the dartos muscle from the scrotum; it is commonly referred to as the dartos layer. In contrast, the deep penile fascia is a denser fascial sheath, known as Buck's fascia, which envelops both corpora cavernosa and splits to envelop the corpus spongiosum (Fig. 76.22B). Distally, it blends with the tunica albuginea covering all three corporal bodies. Proximally, it is continuous with the dartos muscle and the deep perineal fascia. Bleeding from a tear in the corporal bodies is usually contained within Buck's fascia, and ecchymosis is limited to the penile shaft.

Root The root (radix) of the penis consists of three masses of erectile tissue in the urogenital triangle: namely, the two crura and the bulb, firmly attached to the pubic arch and the perineal membrane, respectively. The crura are the posterior extensions of the corpora cavernosa, while the bulb is the dilated posterior end of the corpus spongiosum. The urethra enters the bulb via its posterior surface and travels the length of the penile shaft within the corpus spongiosum.

Shaft The major portion of the penile shaft consists of the paired corpora cavernosa. These contain erectile tissue, enclosed within a dense fibroelastic sheath of connective tissue, the tunica albuginea (see Figs 76.21E, 76.22). The outer longitudinal and inner circular fibres of the tunica form an undulating meshwork when the penis is flaccid but become tightly stretched on erection. The circular fibres surround each corpus separately and the corpora are then surrounded as a unit by the outer longitudinal fibres. Smooth muscle bundles traverse the erectile bodies to form endothelium-lined cavernous sinuses, which give the erectile tissue a spongy appearance on gross examination. Blood flow into the sinuses leads to penile engorgement and compression of venous outflow channels, resulting in penile erection.

The corpora cavernosa are not distinct, separate structures but are divided in the midline by a fibrous septum that is continuous with the deep circular fibres of the tunica albuginea. The septum is complete and thick proximally, but incomplete distally, permitting communication and exchange of blood flow between the corporal bodies. The wide median groove along the ventral aspect of the corpora cavernosa is occupied by the corpus spongiosum, containing the urethra. Dorsally, a similar, but narrower, groove contains the dorsal neurovascular bundle. The corpora cavernosa end distally within the glans penis as individual, rounded cones (see Fig. 76.19D). The corpus spongiosum contains less erectile tissue than the corpora cavernosa, and is enclosed by a thinner layer of tunica albuginea. The urethra traverses the length of the corpus spongiosum, terminating at a slit-like meatus on the tip of the glans penis, which is, itself, an expansion of the corpus spongiosum. Numerous small preputial glands secreting sebaceous smegma line the corona along the base of the glans penis.

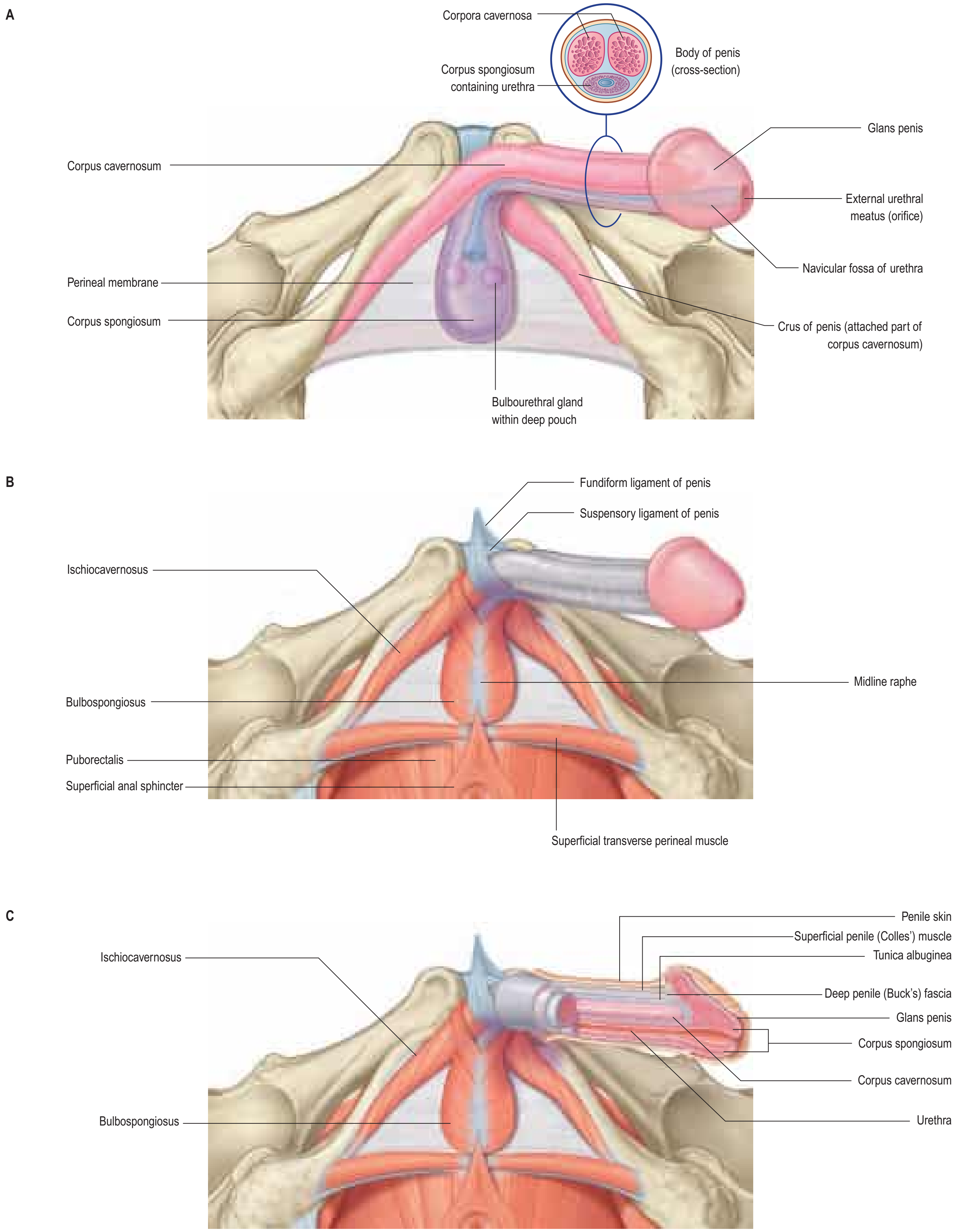

Fig. 76.20 **A**, The erectile tissues of the penis. **B**, The muscles in the superficial perineal pouch. **C**, The muscles and erectile tissues of the penis in section. (**A**–**B**, With permission from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010. **C**, Adapted from Drake RL, Vogl AW, Mitchell A, Tibbitts R, Richardson P (eds), Gray's Atlas of Anatomy, Elsevier, Churchill Livingstone. Copyright 2008.)

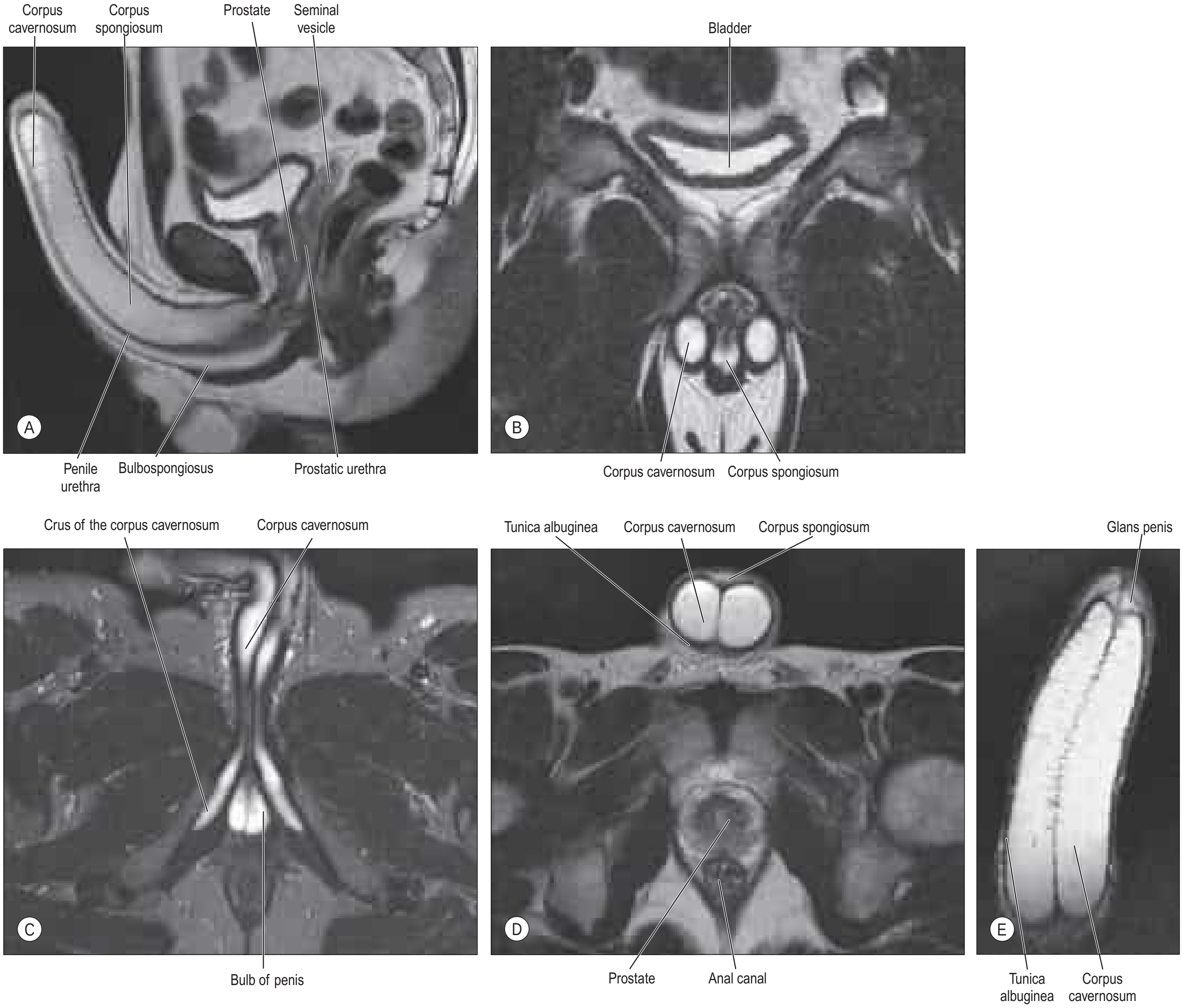

Fig. 76.21 **A–B**, An MRI scan of the penis showing the corpora cavernosum and spongiosum. Note the corpus spongiosum flaring posteriorly into the bulbospongiosus. **C**, An MRI scan showing the bulb of the penis and the attachment of the posterior portion of the corpora cavernosa, the crura. **D–E**, An MRI scan of the penis showing the tunica albuginea.

Vascular supply and lymphatic drainage

Arteries

Blood supply to the corporal bodies is derived from the internal pudendal artery, a branch of the internal iliac artery, which travels within Alcock's canal before reaching the penis and perineum (Fig. 76.23). As it emerges from Alcock's canal, the internal pudendal artery gives off the perineal artery, to supply ischiocavernosus and bulbospongiosus, and the posterior surface of the scrotum, as well as the common penile artery that supplies the deep structures of the penis (Fig. 76.24).

The common penile artery has three main branches: the bulbourethral artery, the dorsal penile artery and the cavernous (deep, cavernosal) artery (Fig. 76.25).

Bulbourethral artery The bulbourethral artery penetrates the perineal membrane to enter the spongiosum from above its posterolateral border and supplies the penile bulb and the urethra, in addition to the corpus spongiosum and glans penis.

Dorsal artery of the penis The dorsal artery of the penis passes between the crus penis and the pubis to reach the dorsal surface of the corporal bodies. It runs alongside the dorsal vein and the dorsal penile nerve, and is attached, together with these structures, to the underside of Buck's fascia. As it courses to the glans penis, it gives off circumferential branches to the corpus spongiosum and urethra. The rich blood supply to the spongiosum allows safe division of the urethra during stricture repair.

Cavernous (deep, cavernosal) artery of the penis The cavernous (deep, cavernosal) artery of the penis is usually a paired vessel that pierces the tunica albuginea of the corpora cavernosa at the hilum of the penis and then travels near the centre of the corporal bodies in the direction of the glans penis. Along its course, it gives off several straight and helicine branches at regular intervals; they open directly into the sinusoidal spaces of the corporal bodies.

Variations in penile vascular anatomy are common, and may include a single or absent cavernous artery or the presence of accessory pudendal arteries (Bare et al 1994, Matin 2006). While penile arterial supply is commonly derived from both accessory and internal pudendal arteries, it may be derived exclusively from either the internal pudendal arteries or the accessory pudendal arteries (Droupy et al 1997). Recognizing such variation is extremely important for any surgeon contemplating penile revascularization surgery.

Veins

Blood leaving the penis is drained by one of three venous systems: superficial, intermediate or deep (see Figs 76.22B, 76.23). The superficial veins are contained within the dartos fascia on the dorsolateral

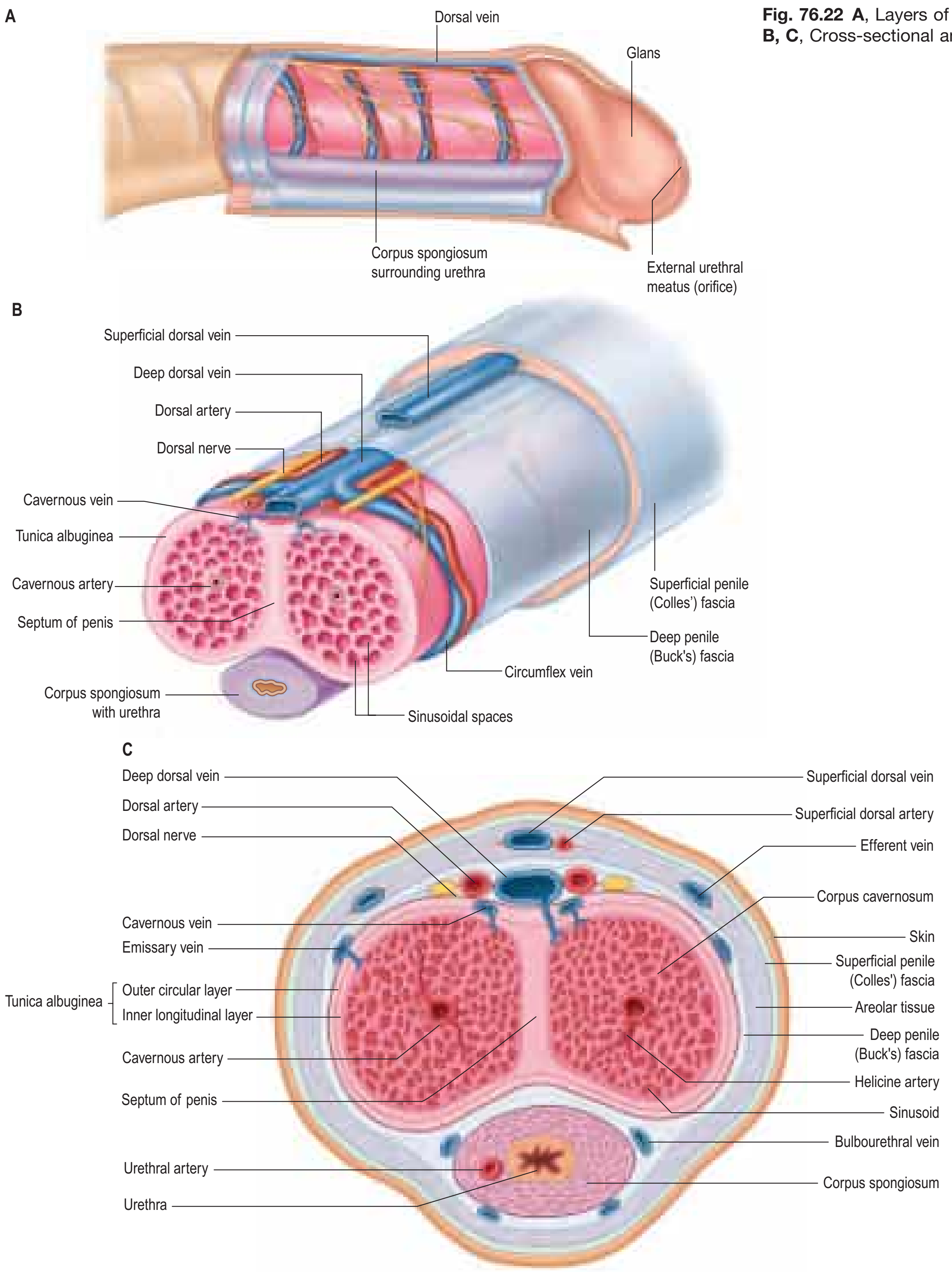

Fig. 76.22 **A**, Layers of the penis and main structures. **B, C**, Cross-sectional anatomy of the penile shaft.

surface of the penis. They receive blood from the penile shaft skin and prepuce, and coalesce at the base of the penis to form a single superficial dorsal vein that drains into the great saphenous vein via the superficial external pudendal veins.

The intermediate drainage system contains the circumflex and deep dorsal veins, which lie within and beneath Buck's fascia. This system receives blood from the glans, corpus spongiosum and the distal two-thirds of the penis. Circumflex veins originate in the corpus spongiosum, and course around the corpora cavernosa to meet the deep dorsal vein perpendicularly. They are only present in the distal two-thirds of the penile shaft, and range between 3 and 10 in number. Intermediate venules in the erectile tissue of the corpora cavernosa arise from the cavernous sinuses and drain into a subtunical capillary plexus. This plexus gives rise to emissary veins that course obliquely through the tunica albuginea, and drain either into the circumflex veins, or directly into the deep dorsal vein. The deep dorsal vein lies in the midline groove between the two corpora cavernosa. It passes inferior to the pubic symphysis at the level of the suspensory ligament, leaving the shaft of the penis at the crus, and drains into the prostatic plexus.

Deep venous drainage occurs via the crural and cavernous (deep, cavernosal) veins, which receive blood from the proximal third of the penis. The emissary veins consolidate into 2–5 veins on the dorsomedial surface of the corpora cavernosa. Crural veins arise in the midline, in the space between the crura, and join the cavernous veins at the hilum of the penis, ultimately draining into the internal pudendal vein.

Lymphatic drainage

The penile and perineal skin is drained by lymphatic vessels that accompany the external pudendal veins to the superficial inguinal nodes. Lymphatics from the glans penis pass to the deep inguinal and external iliac nodes, and those from the erectile tissue and penile urethra pass to the internal iliac nodes.

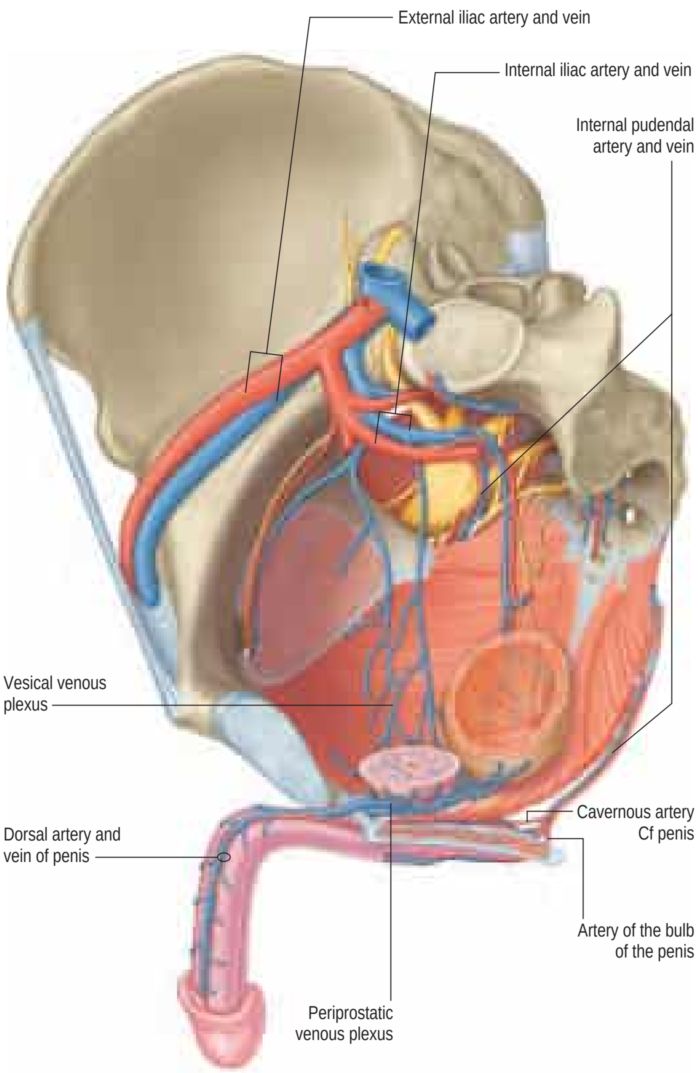

Fig. 76.23 The perineal vessels. (Adapted from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

Innervation

A rich sensory innervation to the glans penis is provided by the dorsal nerve, a division of the pudendal nerve; it follows the course of the dorsal penile arteries (**Fig. 76.26**, see also Fig. 76.2). Small branches from the perineal nerve provide additional innervation to the skin on the ventrum of the penis, as far distally as the glans (Uchio et al 1999).

Sympathetic and parasympathetic input to the corpora cavernosa is provided by the cavernous (deep, cavernosal) nerve, which originates in the pelvic plexus. It courses along the posterolateral aspect of the prostate before exiting the pelvis, together with the urethra, and entering the corpora cavernosa at the crus. Before entering the cavernous bodies, the cavernous nerve sends branches to the corpus spongiosum. Parasympathetic input to the penis (excitatory) originates in the neurones of the first four sacral segments of the spinal cord and travels via the pelvic splanchnic nerves (nervi erigentes) to the pelvic plexus (see Fig. 76.26), where preganglionic fibres synapse on postganglionic neurons that give rise to the cavernous nerve. Sympathetic input to the penis (inhibitory) originates in the intermediolateral columns of the eleventh and twelfth thoracic, and first lumbar segments of the spinal cord, and travels through the sympathetic trunk before descending to the pelvic plexus. Parasympathetic stimulation produces vasodilation, while sympathetic innervation causes vasoconstriction, contraction of the seminal vesicles and prostate, and seminal emission.

Microstructure

The corpora cavernosa are cylinders of spongy erectile tissue, composed of interconnected endothelium-lined sinusoids, separated by smooth muscle trabeculae, and surrounded by elastic fibres, collagen and loose areolar tissue. The sinusoids are larger in the centre. Terminal cavernous nerves and helicine arteries are intimately associated with smooth muscle. The helicine arteries are contracted and tortuous in the flaccid state, and straight and dilated in the erect state.

The tunica albuginea surrounding the corpora consists of an inner circular and outer longitudinal layer of elastic fibres arranged in a latticed network. Venous drainage from the corpora cavernosa originates in the trabeculae between the peripheral sinusoids and the tunica albuginea (the subtunical venous plexus) before exiting the corpora via the emissary veins.

Emissary veins travel between the two layers of the tunica albuginea, and exit the outer layer in an oblique fashion. The outer layer compresses the emissary veins when the penis becomes engorged with blood, preventing venous outflow, and thereby maintains the erection.

The structure of the corpus spongiosum is similar to that of the corpora cavernosa, with the exception of larger sinusoids and a thinner tunica albuginea. The glans has no tunical covering.

ERECTION AND EJACULATION

Penile erection is a neurovascular process, initiated by sexual stimulation and relaxation of the smooth muscle of the corpora cavernosa and mediated by various neurotransmitters produced by the parasympathetic nerves and the endothelium. The most important neurotransmitter is nitric oxide, which acts via the cyclic guanosine monophosphate (cGMP) second messenger system. Following smooth muscle relaxation, there is rapid influx of blood flow from the helicine arteries, filling the cavernous spaces and leading to penile tumescence. The resulting distension compresses the emissary veins, preventing venous outflow. This so-called 'veno-occlusive mechanism' converts tumescence into an erection. Continuing cutaneous stimulation of the glans and frenulum contributes significantly towards maintaining an erection and initiating orgasm and ejaculation. Erection, therefore, depends on a variety of factors, including a normal psychogenic response to stimulation, intact parasympathetic nerves, healthy corporal smooth muscle capable of relaxation, patent arteries capable of delivering blood at the required rate, and a normal venous system. Following ejaculation, the sequence just described is reversed; smooth muscle contraction in response to sympathetic stimulation leads to penile detumescence.

Emission and ejaculation are separate, but related, processes. Emission is the transmission of seminal fluid from the vas deferens, prostate and seminal vesicles into the prostatic urethra and is under sympathetic control. Ejaculation is the expulsion of seminal fluid from the prostatic urethra to the exterior and has both autonomic and somatic components. The first discernible event during ejaculation is contraction of bulbospongiosus, which occurs approximately six times and is under somatic control. Antegrade ejaculation requires concomitant bladder neck closure and periurethral muscle contraction (Ch. 75). Relaxation of the external urethral sphincter, under autonomic control, permits expulsion of seminal fluid from the prostatic to the bulbar urethra, and from there to the exterior. The timing of this process is such that the ejaculate is expelled from the external urethral meatus between the second and final contraction of bulbospongiosus; in some men, ejaculation may be pulsatile.

Erectile dysfunction and priapism

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SCROTUM

The scrotum is composed of multiple layers of tissues, including skin, dartos muscle and external spermatic, cremasteric and internal spermatic fasciae (see Fig. 76.3). The internal spermatic fascia is loosely attached to the parietal layer of the tunica vaginalis.

Scrotal skin is thin, pigmented, devoid of fat, hair-bearing, and rich in sebaceous and sweat glands. It is also richly innervated by sensory nerves that respond to stimulation of the skin and hairs, and to changes in temperature. The appearance of the scrotal skin may vary from smooth to rugated, depending on the degree of contraction of the underlying dartos muscle. A midline raphe extends from the urethral meatus, down the ventral penile shaft and to the anus, signifying the line of fusion of the genital tubercles (see Fig. 72.20). It is a relatively avascular plane and, therefore, a popular location for incisions used to access both scrotal compartments. Deep to the raphe, the scrotum is separated into two compartments by a septum composed of all the

Erectile dysfunction is the inability to achieve tumescence despite adequate sexual stimulation. The pathophysiology of erectile dysfunction is complex, and involves vascular, neurogenic and psychogenic components. Common aetiologies underlying a diagnosis of erectile dysfunction include: inability to relax the cavernous smooth muscle; arterial insufficiency due to atheromatous disease; neurogenic dysfunction due to diabetes or surgery; and radiation therapy.

A prolonged erection that fails to subside after ejaculation is called priapism. Priapism can, broadly, be classified as being high-flow (non-ischaemic) or low-flow (ischaemic). Low-flow priapism, which is due to impaired efflux of venous blood from the penis, is characterized by significant pain and is a medical emergency. If untreated, priapism leads to ischaemia of the corporal smooth muscle and irreversible erectile dysfunction. The aetiology of low-flow priapism includes conditions such as sickle-cell disease and leukaemia, and the use of intracavernosal injection therapy with medications such as prostaglandin. In contrast, high-flow priapism is the consequence of excessive blood flow into the penis, usually secondary to a fistula between the cavernous arteries and sinusoidal spaces.

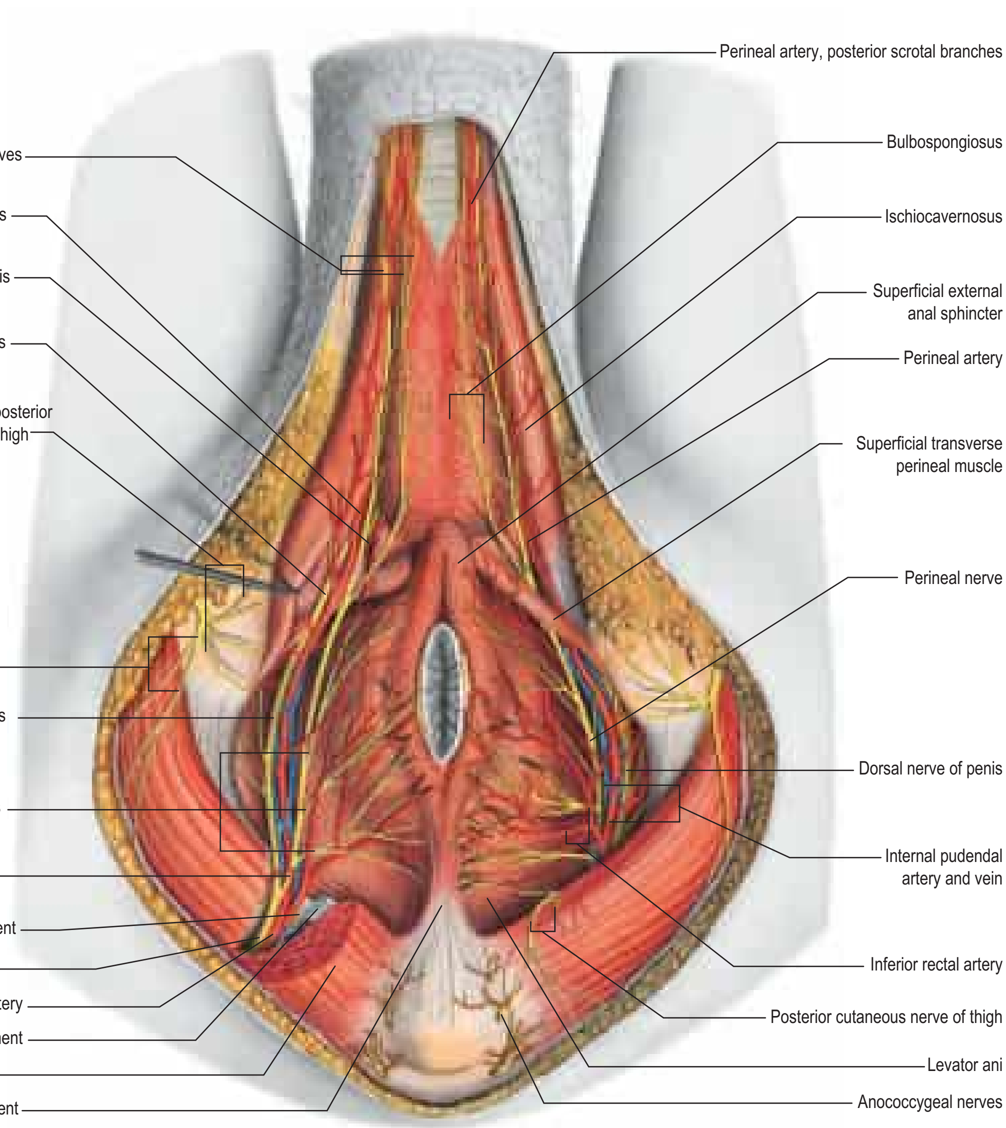

Fig. 76.24 The blood vessels and nerves of the perineal region and external genitalia in the adult male. The fat body of the ischio-anal fossa has been removed and gluteus maximus has been incised in order to expose the course of the pudendal nerve and internal pudendal artery. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban and Fischer. Copyright 2013.)

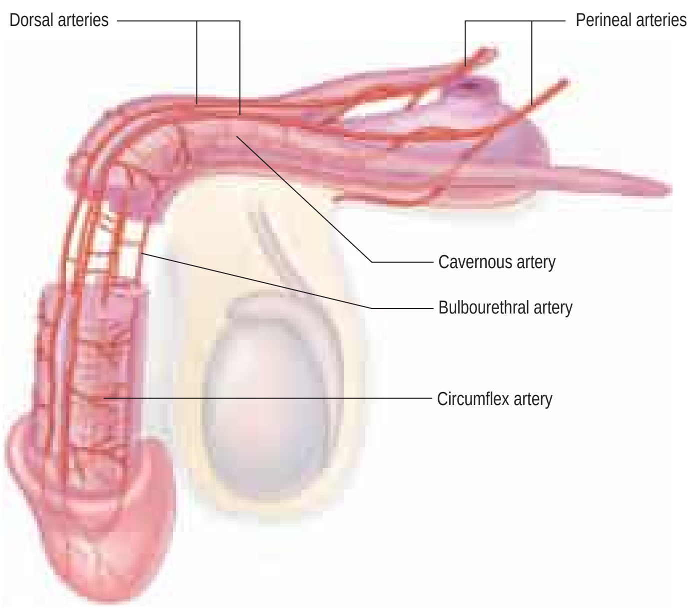

Fig. 76.25 The arterial blood supply to the body of the penis.

layers of the scrotal wall except the skin. The testes are suspended by the spermatic cords within these compartments.

The dartos layer of smooth muscles is continuous with Colles', Scarpa's and the dartos fascia of the penis. The external spermatic, cremasteric and internal spermatic layers of the scrotum are continuous with the corresponding layers in the spermatic cord, and arise from the aponeuroses of external oblique and internal oblique and the transversalis fascia of the abdominal wall, respectively. The testis is also fixed to the scrotal wall at its lower pole by a fibrous band of tissue known as the gubernaculum.

Developmental anomalies of the scrotum Congenital agenesis of the scrotum is a very rare anomaly characterized by the absence of the scrotum; no rugae are noted in the perineum (Yilmaz et al 2013). Penoscrotal transposition, in which the scrotum is located superior and anterior to the penis, is a consequence of abnormal genital tubercle development and is associated with hypospadias, renal agenesis and dysplasia, and imperforate anus.

Vascular supply and lymphatic drainage

Arterial supply to the scrotum includes external pudendal branches of the femoral artery, scrotal branches of the internal pudendal artery, and a cremasteric branch from the inferior epigastric artery. Of note, the anterior scrotal wall is supplied primarily by branches of the external pudendal artery, which run parallel to the rugae and do not cross the raphe; this anatomical detail is important for planning appropriate incisions during scrotal surgery.

The scrotal vessels are arranged in a dense subcutaneous network, which facilitates heat loss and regulation of scrotal temperature. Venous drainage follows the arterial network, and simple arteriovenous anastomoses are common.

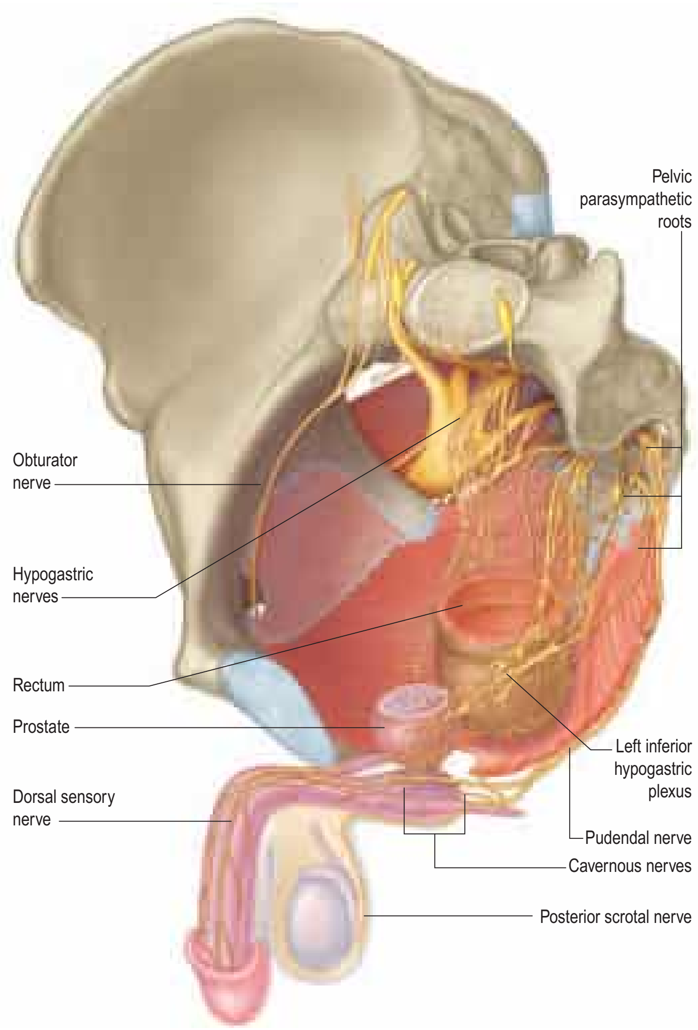

Fig. 76.26 The nerve supply to the penis. The corpus cavernosum of penis receives both a parasympathetic and a sympathetic innervation from the cavernous nerves. It should be noted that, in life, multiple cavernous nerves emanate from the prostatic plexus and intertwine with both dorsal sensory nerves. The afferent fibres from the glans pass via the dorsal nerves of the penis and via the pudendal nerve. (Adapted from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

Scrotal lymphatics do not cross the median raphe. Therefore, lymphatic drainage from the scrotum is always routed to the ipsilateral superficial inguinal nodes.

Innervation

The scrotum is innervated by the ilioinguinal nerve, the genital branch of the genitofemoral nerve, two posterior scrotal branches of the perineal nerve, and the perineal branch of the posterior femoral cutaneous nerve (see Fig. 76.24). Innervation of the anterior third of the scrotum is primarily derived from the first lumbar spinal segment, by way of the ilioinguinal and genitofemoral nerves (see Fig. 76.2). Innervation of the posterior two-thirds of the scrotum is primarily derived from the third sacral spinal segment, via the perineal and posterior femoral cutaneous nerves. In order to anaesthetize the scrotum, spinal anesthesia must, therefore, be injected higher than the L1 level.

Bonus e-book images and table

Fig. 76.1 A, Reference curves for mean testicular volume measured by ultrasound. **B**, Enlargement of the reference curves for mean testicular volume measured by ultrasound.

Fig. 76.12 Gangrene of the appendix testis secondary to torsion in an 8-year-old child.

Table 76.1 Volume of the left and right testis, as measured by ultrasound in boys between infancy and adolescence.

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